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Development of Analysis Strategies for the Detection of Fast Transient Signals from GHz Gravitational Waves

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Abstract

On 14th September 2015, LIGO detected the first gravitational wave signal, marking the beginning of gravitational wave astronomy. To date, all detected gravitational-wave sources have been associated with stellar-origin compact objects. However, some cosmological models predict the existence of exotic sources of gravitational waves which could be in the sub-solar mass range. In order to search for these sources, the GHz regime must be probed.

In recent years, the idea of cavity-based searches for high-frequency gravitational waves has gained significant traction. GravNet is also a cavity-based experiment working on the principle of the inverse Gertsenshtein effect.

Gravitational wave experiments usually rely on the matched filtering technique to analyze the noisy data. The technique requires a template blank of an expected signal waveform in order to detect the signal from noisy data.

Unlike stellar-mass gravitational-wave sources, exotic sources are not yet sufficiently characterized, and their signal waveforms are therefore not well modeled. As a result, extracting weak signals from noisy data is a major challenge in this field.

In this thesis, machine learning based solutions for detecting gravitational wave signals using frequency-domain data are explored. An autoencoder was developed for anomaly detection and trained using observed pure noise data from the SUPAX cavity at Mainz. Moreover, a multi-view classifier was also developed to further classify the data into pure noise and signal-containing samples.

High-frequency gravitational wave signals were then simulated to test the performance of these machine learning based detection pipelines. The autoencoder and classifier were evaluated on the basis of various criteria, including detection efficiency and robustness against noise fluctuations. Finally, their performance was compared with matched filtering and time-domain based machine learning detection pipelines.

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1. Introduction

Humans are fundamentally curious beings. Mankind has spent centuries wondering about the universe and its origins. Countless theories have been proposed about the working of the cosmos and the four fundamental forces governing it. Some were later disproved, while a few paved the way for the advancement of human civilization. One of the most revolutionizing theories in Physics was first published at the beginning of the 20th century by Albert Einstein.

In 1916, the Theory of General Relativity redefined gravity as the manifestation of spacetime curvature caused by mass and energy[1]. Along with its many groundbreaking implications, it also predicted the existence of Gravitational Waves (GWs)[2]. Nearly a Century later, on 14th September 2015, the Laser Interferometer Gravitational-Wave Observatory (LIGO) detected the first Gravitational waves in the history of mankind[3].

Since then, numerous GW wave events have been detected by the LIGO-VIRGO-KAGRA (LVK) collaboration. These detection events include three main categories of GW sources: 1) Binary Black hole merger, 2) Binary Neutron star merger, and 3) Neutron Star-Black hole merger. All of the GW sources detected by LVK to date are consistent with compact binaries of stellar origin, with component masses in the range of ~ 0.2–$200\,M_{\odot}$[4] [5].

Currently, within the modern ΛCDM model framework, theories predict the GW sources outside of the stellar mass range, such as supermassive black holes and primordial black holes[6]. To detect the exotic sources of GWs, we must probe the nHz and GHz ranges. As the LVK is mainly sensitive in the 10 Hz to a few kHz frequency range, exploring these complementary regimes requires alternative detector technologies. Particularly in the High Frequency Gravitational waves (HFGW) regime, Radio Frequency (RF) cavity based experiments have recently emerged as promising candidates. These HFGW experiments can be based on the inverse Gertsenshtein effect, where GWs convert into Electromagnetic (EM) waves in the presence of a strong magnetic field. The induced magnetic field is then amplified within the microwave cavity, and the resulting radio-frequency power can then be measured[7].

The biggest challenge in HFGW experiments is detecting transient signals from the noisy data. Unlike the stellar mass GW sources, potential exotic sources of GW are not yet well characterized. Hence, expected signals cannot be predicted with accuracy in advance. This uncertainty in the signal waveform limits the abilities of the widely used matched-filtering technique for signal detection and analysis, as matched filtering requires an expected signal template to extract a signal from noisy data.

In recent years, the idea of using Machine Learning (ML) tools for GW detection has gained significant traction. Machine learning methods can identify data patterns and could

be used to characterize noise in GW datasets. Moreover, unlike traditional template-based approaches such as matched filtering, many ML-based approaches do not require an explicitly modeled waveform template and can therefore be applied in weakly modeled or unmodeled search scenarios. This can enable faster and more robust HFGW detection.

In this thesis, we will explore ML based GW signal detection approach using neural networks. In particular, we will use an Auto-Encoder (AE) for anomaly detection. The AE is trained on the noise only data from the SUPAX cavity at Mainz, allowing the AE to learn to reconstruct the underlying noise patterns. After training, simulated high-frequency gravitational wave (HFGW) signals are injected into the test dataset in order to evaluate the detection pipeline. Since the AE has only learned the noise distribution during training, it fails to accurately reconstruct the injected signal waveforms and instead reconstructs only the learned noise patterns. This results in an increased reconstruction loss, which can be used to define a threshold for identifying anomalous events and thereby detecting potential GW signals.

In addition to the anomaly-detection approach based on the AE, we also developed a supervised machine-learning classifier. This classifier is trained using both noise-only data and noise combined with simulated GW signals, together with the corresponding reconstructed outputs and reconstruction-loss values produced by the AE. The overall objective of this combined framework is to improve the reliability and sensitivity of gravitational wave signal identification by leveraging both unsupervised anomaly detection and supervised classification methods within a unified detection pipeline.

Finally, we will test the detection pipelines and compare them with matched filtering and excess power method to evaluate the ML-based detection strategies.

2. Theory

2.1. General Relativity

In 1916, Albert Einstein published the Theory of General Relativity, revolutionizing our understanding of gravity. It described gravity as a geometric property of a curved spacetime and not as a Newtonian force. The curvature is a result of the distribution of mass and energy. Test particles follow geodesic trajectories determined by this curvature when they are not acted upon by any non-gravitational forces. Mathematically, the relationship between the mass–energy distribution and the curvature of spacetime is given by Einstein's field equations. This curvature is determined locally by the energy–momentum content, represented by the tensor $T_{\mu\nu}$.
The Einstein Field Equation reads as follows:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}\,. \tag{2.1}$$

The left-hand side of the equation can be described by the Einstein tensor, denoted by $G_{\mu\nu}$. The Einstein tensor is a geometric quantity that summarizes how the spacetime is curved. It consists of the Ricci curvature tensor represented by $R_{\mu\nu}$, which measures the degree to which the geometry of a given metric tensor differs locally from that of Minkowski spacetime η = diag(-1,+1,+1,+1).

The Ricci curvature tensor can be obtained by contracting the Riemann tensor ($R^{\alpha}{}_{\beta\mu\nu}$) in the first and third indices. The Riemann tensor is given by,

$$R^{\alpha}{}_{\beta\mu\nu} = \partial_\mu\Gamma^{\alpha}{}_{\beta\nu} - \partial_\nu\Gamma^{\alpha}{}_{\beta\mu} + \Gamma^{\alpha}{}_{\gamma\mu}\Gamma^{\gamma}{}_{\beta\nu} - \Gamma^{\alpha}{}_{\gamma\nu}\Gamma^{\gamma}{}_{\beta\mu}\,. \tag{2.2}$$

Where $\Gamma^{\gamma}{}_{\beta\nu}$ is called a Christoffel symbol and it can be described as:

$$\Gamma^{\alpha}{}_{\mu\nu} = \frac{1}{2}g^{\alpha\beta}\left(\partial_\mu g_{\beta\nu} + \partial_\nu g_{\mu\beta} - \partial_\beta g_{\mu\nu}\right)\,. \tag{2.3}$$

Where ∂_μ indicates the partial derivative $\frac{\partial}{\partial x^\mu}$.

On the other hand, Ricci scalar (R) can be derived by contracting the Ricci tensor as follows: $R = g^{\mu\nu}R_{\mu\nu}$ whereas Λ is the cosmological constant. It can be interpreted as the uniform vacuum energy density. When Λ is set to a positive value, it behaves as a repulsive contribution to the gravity and leads to the accelerated expansion of the universe. In the ΛCDM model, this vacuum energy is often associated with dark energy.[8]

In this thesis, we will use G=c=1, so that energy, time, and length are expressed in

units of mass.

2.2. Linearized Theory of Gravitational Waves

Gravity is considered a weak force because at the particle scale it is 10^{36} times weaker than the Electromagnetic force. Despite this, Gravity dominates on Astronomical scales as it is always attractive and has an infinite range. Gravitational field strength is strongest near compact massive objects like Black holes or Neutron Stars. However, it is safe to assume that the gravitational fields are sufficiently weak in most regions of the universe. This allows us to treat spacetime curvature as a small perturbation of flat (Minkowski) spacetime. This is known as Linearized Gravity, where spacetime metric $g_{\mu\nu}$ can be defined as

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \tag{2.4}$$

Where $\eta_{\mu\nu}$ is the flat Minkowski metric and $h_{\mu\nu}$ is the small perturbation metric.

2.3. Inverse Gertsenshtein effect

The inverse Gertsenshtein effect describes the process by which gravitational waves convert into electromagnetic radiation in the presence of a background magnetic field [9]. To obtain the accurate EM wave function, generated by the inverse Gertsenshtein effect, we will begin by using Maxwell's equations in the curved spacetime:

$$\nabla_\nu F^{\mu\nu} = \frac{j^\mu_{\text{magnet}}}{\sqrt{-g}}, \nabla_\nu F_{\alpha\beta} + \nabla_\alpha F_{\beta\nu} + \nabla_\beta F_{\nu\alpha} = 0, \tag{2.5}$$

where $g_{\mu\nu}$ is the metric tensor and $g = det(g_{\mu\nu})$. By using metric perturbation theory, we can write it as $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$

Similarly, the electromagnetic strength tensor $F_{\mu\nu}$ can be expressed as, $F_{\mu\nu} = \bar{F}_{\mu\nu} + F^h_{\mu\nu}$. $\bar{F}_{\mu\nu}$ is the EM field in flat spacetime. In our case, it arises due to the electromagnet placed around the cavity. Thus, it is induced by the current j^μ_{magnet}, and we have,

$$\partial_\nu \bar{F}^{\mu\nu} = j^\mu_{\text{magnet}}. \tag{2.6}$$

While $F^h_{\mu\nu}$ represents the induced EM field that satisfies Maxwell's equations with the first-order Gravitational waves perturbation.

$$\partial_\nu F^{\mu\nu}_h = \partial_\nu(-\frac{h}{2}\,\bar{F}^{\mu\nu} - \bar{F}^{\nu\alpha}h^\mu{}_\alpha + \bar{F}^{\mu\beta}h^\nu{}_\beta) \equiv j^\mu_{\text{eff}} \tag{2.7}$$

$$\partial_\nu F^h_{\alpha\beta} + \partial_\alpha F^h_{\beta\nu} + \partial_\beta F^h_{\nu\alpha} = 0, \tag{2.8}$$

where j^{μ}_{eff} is the effective current produced by the interaction between GW and background EM field. This current acts as a source term in the inhomogeneous Maxwell's equation. Equation 2.8 is satisfied as long as, $F^{h}_{\mu\nu} = \partial_{\mu}A^{h}_{\nu} - \partial_{\nu}A^{h}_{\mu}$. For given background static magnetic field, $j^{0}_{\text{eff}} = 0$ and as a result $A^{0}_{h} = 0$. By substituting this definition in 2.3 we get,

$$\Box A^{(h)}_{\mu} = -\, j^{\text{eff}}_{\mu}. \tag{2.9}$$

Equation 2.9 holds as long as the back conversion of the induced EM field to GW is negligible.

For a plane GW with wave vector $k = \omega(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, where θ is the polar angle measured from the z axis and the ϕ represents the azimuthal angle measured in the xy plane from the x axis. A convenient choice of two transverse unit vectors is $v = (-\sin\phi, \cos\phi, 0)$ and $u = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta)$.

The GW in the transverse traceless (TT) frame can be written as

$$h^{\text{TT}}_{ij} = \left(h_{+}e^{+}_{ij} + h_{\times}e^{\times}_{ij}\right) e^{-i(\omega t - \mathbf{k}\cdot\mathbf{r})}, \tag{2.10}$$

where e^{+}_{ij} and $e^{\times}_{ij}$ are respectively the usual + and × polarization tensors. while, h^{TT}_{ij} denotes the gravitational-wave strain tensor in the transverse-traceless gauge, while h_{+} and $h_{\times}$ are the amplitudes of the plus and cross polarization modes, respectively.
The resulting effective current can be written as

$$\mathbf{j}_{\text{eff}} = \sum_{\lambda=+,\times} h_{\lambda}\,(i\mathbf{k} + \nabla) \times \left(e^{\lambda}\mathbf{B}_{0}\right) e^{-i(\omega t - \mathbf{k}\cdot\mathbf{r})}, \tag{2.11}$$

where B represents the static magnetic field and $(e^{\lambda}\mathbf{B}_{0})_{i} = e^{\lambda}_{ij}B^{j}$.

2.4. Detectors

Advanced Michelson interferometers like LIGO and VIRGO have proven to be highly successful in gravitational wave detection. Another way of searching for gravitational waves is by probing their coupling with electromagnetic fields.

According to the inverse Gertsenshtein effect, a gravitational wave can couple directly to an electromagnetic field. In the presence of a static external magnetic field, the incident GW induces an electromagnetic wave oscillating at the same frequency. This induced EM field can be resonantly enhanced using Microwave cavities and the resulting radio frequency power can be detected.[7]

The signal power P_{sig} can be calculated in the proper detector frame as[7]

$$P_{\text{sig}} = \frac{1}{2}\,Q\,\zeta^{3}\,g\,V^{5/3}_{\text{cav}}\,(\omega_{n}h_{0}B_{0})^{2}\;\frac{1}{\mu_{0}c^{2}}, \tag{2.12}$$

where,

- Q is the quality factor,
- ω_g is the frequency of GW,
- V is the volume of cavity,
- h_0 is the strain of GW strain,
- B_0 is the external magnetic field.
- η_n is the dimensionless coupling constant.

The noise data used in this thesis was obtained from the measurements of the SUPAX cavity at Mainz.[10]. SUPAX is aimed to detect dark photons and axion-like particles with approximate mass of 34μeV.

The SUPAX cavity has inner dimensions of $150 \times 22.8 \times 30\ mm^3$, while the corners are rounded (r = 9 mm) to minimize the boundary effects. These dimensions are carefully chosen to maximize the cavity volume and to fit the cavity inside the cryostat of 52 mm diameter and to keep the mode next to the TM_{010} more than 100 MHz away.

Based on the dimensions, the resonance frequency of the cavity turns out to be f =8.47 GHz at the room temperature.

The unloaded quality factor of the cavity is measured to be $Q_0 = 40726 \pm 4250$. This is in agreement with the simulated quality factor of $Q - 0 = 40059 \pm 128$[11].

2.5. High Frequency Gravitational Waves Simulation

To evaluate the anomaly detection capability of the designed AE, we simulated High-Frequency Gravitational Wave signals.

For simplicity, we consider only one type of signal, a (transient) chirp sourced by a Primordial Black Hole (PBH) merger event. To further simplify the calculations, we assume the system is an equal mass binary.

Since the detection bandwidth of GravNet's cavity is limited to 0.441 MHz, only a small portion of the chirp signal is expected to be observed. As a result, the merger and ringdown phases are unlikely to be seen by the cavity in most cases, and the observable signal will predominantly correspond to the linear inspiral segment. Therefore, for the purpose of testing the detection pipeline in this thesis, we restrict our simulations to the linear chirp segment of the signal.

The linear chirp signal can be described by

$$S(t) = Ae^{i\phi(t)}. \tag{2.13}$$

Equation 2.13 is the complex analytic form of the linear chirp signal where,A represents the amplitude of the signal[1] and ϕ is the instantaneous phase of the linear chirp, which can be written as

$$\phi(t) = 2\pi \left(f_0 t + \frac{1}{2}\alpha t^2 \right), \tag{2.14}$$

where the f_0 is the starting frequency and α is the sweep rate which can be written as

$$\alpha = \frac{f_e - f_0}{T}. \tag{2.15}$$

Here, f_e is the end frequency and T is the time it takes to sweep from f_0 to f_e. The time derivative of ϕ determines the instantaneous frequency. It can be expressed as

$$f = \frac{1}{2\pi}\frac{d\phi(t)}{dt} = \alpha t + f_0. \tag{2.16}$$

The sweep rate α is proportional to the PBH mass. As we assume both PBHs to have the same masses, α can be calculated as

$$\alpha = \frac{96}{5 \cdot 2^{1/3}} \pi^{8/3} \left(\frac{Gm}{c^3} \right)^{5/3} f^{11/3}. \tag{2.17}$$

In this thesis, we consider PBH masses in the range of $10^{-13} M_\odot$ to $10^{-10} M_\odot$ for the signal simulation. The resulting sweep rates and signal duration are given in Table 2.1. Please note that, the frequency range of most of the simulated signals lies between f_0=8.494 GHz to f_e=8.503 GHz, corresponding to approximately four times the Full Width Half Maximum (FWHM) (from 8.4981 GHz to 8.4985 GHz) of the cavity. The only exception is the signals corresponding to the PBH mass of $3.162 \times 10^{-11} M_\odot$; these signals have a frequency range spanning two times the FWHM, with $f_0 = 8.4959$ GHz and $f_e = 8.5007$ GHz. This decision was made to keep the signal duration short and allow more test signals to be included within a single noise file. The signal starts and ends far from the FWHM bandwidth because, in practice, the inspiral phase of the merger extends over a much broader frequency range; this is done to mimic that behavior.

Figure 2.1 shows the linear chirp signal.

[1]For simplification, A is considered to be constant across all frequencies.

Mass ($M_\odot$)	Sweep Rate (s^{-2})	Signal Duration (s)
10^{-13}	2.439×10^{8}	3.799×10^{-2}
3.162×10^{-13}	1.662×10^{9}	5.576×10^{-3}
10^{-12}	1.132×10^{10}	8.184×10^{-4}
3.162×10^{-12}	7.712×10^{10}	6.292×10^{-5}
10^{-11}	5.25×10^{11}	1.763×10^{-5}
3.162×10^{-11}	3.58×10^{12}	2.588×10^{-6}
10^{-10}	2.439×10^{13}	3.799×10^{-7}

Table 2.1.: Sweep rate and signal duration for given PBH mass

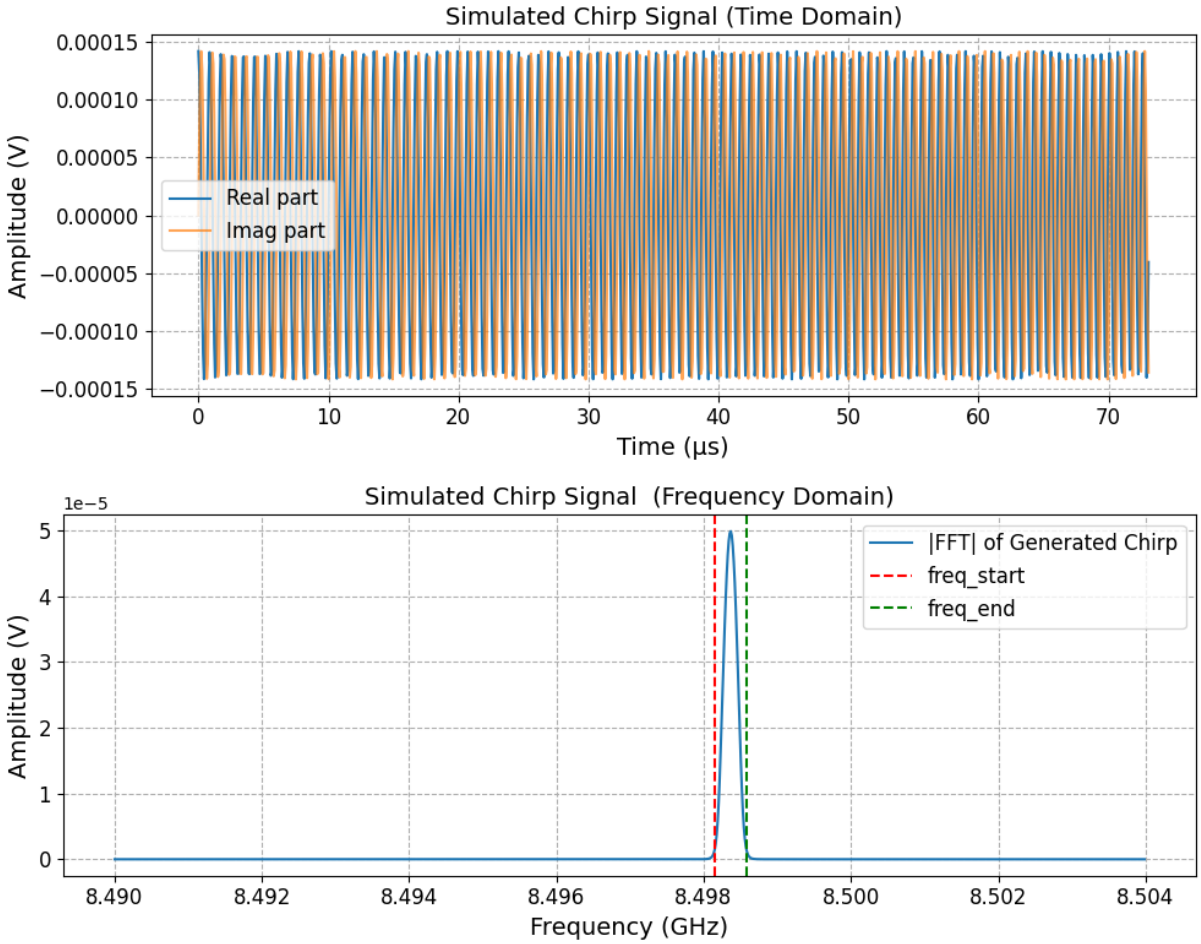


Figure 2.1.: Simulated linear chirp signal of inspiral phase of equal mass PBH binary merger. First image shows the linear chirp in time domain, while the second image shows the linear chirp in the frequency domain. The sharp peak in the frequency domain arises because the plotted spectrum is limited by the cavity FWHM, so only a small portion of the chirp spectrum is shown rather than the full frequency range of the sweep. The dotted lines in the second image are representing the FWHM bandwidth of the cavity.

In practice, the signal is expected to have a resonance component introduced by the RF cavity used for the measurements. To mimic the signal as observed by the cavity, we will take into account the resonance response of the cavity indicated by L(f). The

The cavity's resonance response can be expressed in the form of the Breit-Wigner function [12] as

$$L(\omega) = \frac{1}{(\omega^2 - \omega_0^2)^2 + \gamma^2\omega_0^2}, \tag{2.18}$$

where ω_0 represents the resonance frequency and γ represents the damping width. For simplicity, we will use Full-Width-Half-Maximum (FWHM) as the damping width[2]. The experimentally observable signal is the convolution of the chirp signal with the resonance response in the time domain. From Eq. 2.13 and 2.18 we can write,

$$(S * L)(t) = \int_{-\infty}^{\infty} S(\tau)\, L(t-\tau)\, d\tau \tag{2.19}$$

After the signal is generated and multiplied with the resonance response of the cavity in the frequency domain, it is added to the noise data obtained from the SUPAX cavity at Mainz [11].

2.5.1. Signal injection

To test the detection pipeline using these simulated signals, we first inject them into the noise data. The noise data from the SUPAX cavity in Mainz is in the IQ format. As the name suggests, IQ data has two components: the in-phase component, I, and the quadrature component, Q. These components describe the signal relative to two reference waves that are 90° out of phase.

Together, they form a complex-valued representation of the data, $x(t) = I(t) + iQ(t)$, which preserves both amplitude and phase information. This makes IQ data especially useful for analyzing modulated, oscillatory, or narrowband signals in the time domain.

After the signal is generated, it is Fourier transformed into the frequency domain and multiplied by the cavity resonance response. The modified signal is then transformed back to the time domain before being injected into the noise data.

The detection pipeline used in this thesis processes input vectors of length 1024 data points, where each vector corresponds to a duration of $7.31, \mu$s. Simulated signals could either be shorter or longer in duration compared to the input vectors. The signal injection process differs for these two cases. In the case of signals with a duration of $7.31\,\mu$s or less, the signal is injected into every input vector. The starting point of the signal is randomly selected within the input vector, with the constraint that the complete simulated signal remains contained within the 1024-sample window. Even if the signal's duration is shorter than $7.31\,\mu$s, it is not repeated within a given 1024-sample window.

[2] FWHM $= \sqrt{\omega_0^2 + \gamma\omega_0} - \sqrt{\omega_0^2 - \gamma\omega_0}$, but for simplicity we set $\gamma \equiv$ FWHM.

With this signal injection strategy, we obtain one simulated signal per input vector. Resulting in 109375 simulated signals in one noise file of 8 s duration.

On the other hand, for signals with a longer duration than 7.31 μs, the simulated signal is injected in the full noise file of length 8 s, sequentially and repeatedly. As a result, only 9775 signals are added per file for the PBH mass of $10^{-12}M_{\odot}$, 1434 signals for PBH mass of $3.162 \times 10^{-13}M_{\odot}$, while PBH mass of $10^{-13}M_{\odot}$ has 211 signals per file.

Following this, the signal-to-noise ratio (SNR) of each simulated signal is defined as the ratio between the root mean square (RMS) amplitude of the signal and the standard deviation of the corresponding noise data (σ_{noise}). The standard deviation is computed from the whole data file of 8 s and its value is 2.688×10^{-5} The SNR is therefore given by

$$\text{SNR} = \frac{\text{RMS}_{\text{signal}}}{\sigma_{\text{noise}}}. \tag{2.20}$$

The asteroid-mass window ranges from component masses of $10^{-16}M_{\odot}$ to $10^{-10}M_{\odot}$, where all dark matter could be composed of PBHs[13]. However, for testing purposes, only the PBH mass range from $10^{-13}M_{\odot}$ to $10^{-10}M_{\odot}$ was simulated. The lower limit on this component mass range is due to the limited availability of data during this thesis. As seen in the table 2.1, signal duration increases with decreasing PBH mass, and for $10^{-14}M_{\odot}$ component mass, the signal duration is 1.763 s. Thus, it was not possible to simulate a sufficient number of signals of this duration to achieve statistical significance during testing.

All of the signal injection process takes place in the time domain, and later, for testing purposes, the FFT is performed on the input vectors of 1024 data points in length. The details about the pre-processing of data for the detection pipeline are given in the section 3.3.2

2.6. Machine Learning

Machine learning is a branch of Artificial intelligence that focuses on the development and study of Statistical Algorithms that learns from given data and improve their performance. Such algorithms can be used to identify patterns, predict trends, and, in the context of this thesis, detect anomalous behavior in the data. Precisely, Anomaly detection is a process of identifying patterns that do not follow an expected or well-defined interception of normal data.[14] In terms of machine learning based anomaly detection, it is simply the detection of new or unknown data/patterns that the system is not aware of during training. [15]

In this thesis, we will use Auto-Encoders (AE) as Anomaly detectors.

2.6.1. Autoencoder

An Autoencoder is a type of Neural Network designed to encode the input into a compressed latent representation and then decode it to produce a reconstruction that is as close as

possible to the original input.[16]
In general, an Autoencoder consists of two main components: an encoder and a decoder. The encoder maps the input data to a latent representation, while the decoder maps this latent representation back to the original input space.

Let the input sample be denoted by $x \in \mathbb{R}^n$, where n is the dimensionality of the input space. The encoder (E) and decoder (D) can be represented as $E : \mathbb{R}^n \to \mathbb{R}^p, D : \mathbb{R}^p \to \mathbb{R}^n$, respectively. Here, p is the dimensionality of the latent space. The latent representation can be written as

$$z = E(x), \tag{2.21}$$

and the corresponding reconstruction output can be expressed as

$$\hat{x} = D(z) = D(E(x)). \tag{2.22}$$

Training an autoencoder means finding encoder and decoder functions that minimize the reconstruction loss:

$$\arg\min_{E,D} \ \langle \Delta(x,\ (D \circ E)(x)) \rangle \,. \tag{2.23}$$

Here, Δ denotes the reconstruction loss, i.e., the difference between input and output of the autoencoder[17]. A common choice for the reconstruction loss is the Mean Squared Error (MSE).

For a given dataset of $X = \{x_1, x_2, \ldots, x_N\}$, where N represents the number of samples, the MSE reconstruction can be defined as

$$\mathcal{L}(x_i, \hat{x}_i) = \frac{1}{n} \sum_{j=1}^{n} (x_{ij} - \hat{x}_{ij})^2 \,. \tag{2.24}$$

Hence, the complete training objective can be written as,

$$\min_{E,D} \frac{1}{N} \sum_{i=1}^{N} \frac{1}{n} \sum_{j=1}^{n} (x_{ij} - D(E(x_i))_j)^2 \,. \tag{2.25}$$

The purpose of an Autoencoder is not to copy the input at the output, but to learn the underlying structure of the data through the latent representation. To prevent the identity mapping, bottlenecks are used where $p < n$, or regularization is added to the model. These constraints force the network to preserve the dominant features of the training data instead of simply memorizing each input [16].

In this thesis, a 1-dimensional Convolutional Autoencoder is used. A Convolutional Neural Network (CNN) is a neural network that uses convolutional layers to extract the local patterns from structured data. Instead of connecting every input to every neuron, CNNs use shared filters, which helps them learn spatial or local features efficiently. Usually, CNNs are used on 2D data such as images[18].

A 1D CNN applies convolutional filters along the spatial or temporal dimension. CNNs

are effective at learning local patterns in structured data, making them suitable for capturing noise-related features and localized variations in the input. A simplified 1D convolution operation can be written as

$$h_t = \sigma\left(b + \sum_{r=0}^{k-1} w_r x_{t+r}\right), \tag{2.26}$$

where h_t is the output feature at position t and k is the kernel size, w_r are the convolutional weights, b is the bias term and $\sigma(\cdot)$ is the non-linear activation function[18].

We combined the convolution layers with the autoencoder principle to create a 'CNN autoencoder'. The detailed architecture of this model is given in the section 3.3.2. Convolutional autoencoders preserve the reconstruction-based property of standard autoencoders while learning local dependencies through convolutional filters. Since the same filters are applied across the input, the model captures recurring local patterns in the data. This property makes CNN autoencoders particularly useful for anomaly detection. When such a model is trained on 'normal data', it learns to reconstruct the samples that follow the normal distribution. On the other hand, the samples with anomalous data differ from this learned structure and therefore produce a larger reconstruction loss.

2.6.2. Classifier

The term classification refers to the process in which each input vector is assigned to one of a finite number of discrete categories. In supervised classification, the aim is to learn the mapping between input variables and a categorical output variable using labeled training data [19]. In the simplified case of a binary classifier, there are only 2 classes[20]. Rather than producing a hard label, the probabilistic binary classifiers estimate the conditional probability that a sample belongs to the positive class. Usually, the sigmoid function is used in the last layer of the classifier to obtain the positive class probability. It maps the real-value scores to the interval (0,1), and it can be written as

$$\sigma(z) = \frac{1}{1 + e^{-z}}. \tag{2.27}$$

Equation 2.27 gives the estimated probability of the positive class, and z can be expressed as the inverse of the logistic sigmoid,

$$z = \ln\left(\frac{\sigma}{1 - \sigma}\right). \tag{2.28}$$

Equation 2.28 is known as the logit function [20]. A decision threshold can be set on the output of equation 2.27 to obtain the class labels. We will discuss this in detail in the section 3.3.4.

When different views of the same sample are provided to the classifier with different feature representations, this is referred to as multi-view learning, a learning setting in which

multiple representations of the same instance are used jointly to exploit complementary information for a prediction task [21]. In this thesis, we developed a binary classifier with multi-view feature input.

3. Methods of signal detection

3.1. Time domain

In this section, we will explore the signal detection methods that operate in the time domain.

3.1.1. Matched-Filtering

Matched filtering is the optimal linear filtering technique for detecting a known signal waveform embedded in stationary Gaussian noise[22]. It is one of the most commonly used signal detection techniques in the field of Gravitational Waves.

Let $x(t)$ be our detector output in the time domain, which has a noise component $n(t)$ and might also have a signal component $h(t)$. Assume a linear filter with an impulse response of $q(t)$, the filter response can be written as

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, q(t-\tau)\, d\tau. \tag{3.1}$$

For the exact detection time t_0, we can write the equation 3.1 as

$$y(t_0) = \int_{-\infty}^{\infty} x(\tau)\, q(t_0-\tau)\, d\tau. \tag{3.2}$$

Now, in frequency domain, using Fourier transforms $x(t)$ and $q(t)$, equation 3.2 can be written as

$$\tilde{x}(f) = \int_{-\infty}^{\infty} x(t)\, e^{-2\pi i f t}\, dt, \tag{3.3}$$

$$\tilde{q}(f) = \int_{-\infty}^{\infty} q(t)\, e^{-2\pi i f t}\, dt. \tag{3.4}$$

The filter output in the frequency domain is defined as

$$y(t_0) = \int_{-\infty}^{\infty} \tilde{x}(f)\, \tilde{q}^*(f)\, e^{2\pi i f t_0}\, df. \tag{3.5}$$

From equation 3.5, the expected signal contribution can be written as

$$\mu = \mathbb{E}[y(t_0)] = \int_{-\infty}^{\infty} \tilde{h}(f)\, \tilde{q}^*(f)\, e^{2\pi i f t_0}\, df. \tag{3.6}$$

Assuming stationary Gaussian noise and one-sided power spectral density, $S_n(f)$, the variance of the noise contribution at the output is:

$$\sigma^2 = \int_0^\infty S_n(f)\ |\tilde{q}(f)|^2\ df. \tag{3.7}$$

Thus, the signal-to-Noise Ratio (SNR) is

$$\rho = \frac{\mu}{\sigma}. \tag{3.8}$$

From equations 3.6, 3.7 and 3.8

$$\tilde{q}(f) \propto \frac{\tilde{h}(f)e^{2\pi i f t_0}}{S_n(f)}. \tag{3.9}$$

This is called the Matched Filter transfer function. It is a linear filter that maximizes ρ. It can also be expressed as

$$\tilde{q}(f) = \mathcal{N}\frac{\tilde{h}(f)e^{2\pi i f t_0}}{S_n(f)}. \tag{3.10}$$

where $\mathcal{N}$ is a normalization constant.

Substituting 3.5 into 3.10 gives

$$z(t_0) = 4\,\mathrm{Re}\int_0^\infty \frac{\tilde{x}(f)\tilde{h}^*(f)}{S_n(f)}e^{2\pi i f t_0}\,df. \tag{3.11}$$

The optimal SNR is

$$\rho(t_0) = \frac{(x|h)}{\sqrt{(h|h)}}, \tag{3.12}$$

where the noise-weighted inner product is defined by

$$(a|b) = 4\,\mathrm{Re}\int_0^\infty \frac{\tilde{a}(f)\tilde{b}^*(f)}{S_n(f)}\,df. \tag{3.13}$$

In this case

$$(x|h) = 4\,\mathrm{Re}\int_0^\infty \frac{\tilde{x}(f)\tilde{h}^*(f)}{S_n(f)}\,df \tag{3.14}$$

and

$$(h|h) = 4\,\mathrm{Re}\int_0^\infty \frac{|\tilde{h}(f)|^2}{S_n(f)}\,df. \tag{3.15}$$

To summarize the matched filtering process is,

$$x(t) \xrightarrow{\mathcal{F}} \tilde{x}(f) \tag{3.16}$$

$$\tilde{q}(f) = \frac{\tilde{h}^*(f)}{S_n(f)} \tag{3.17}$$

$$\tilde{z}(f) = \tilde{x}(f)\tilde{q}(f) \tag{3.18}$$

$$z(t) = \mathcal{F}^{-1}\left[\tilde{x}(f)\frac{\tilde{h}^*(f)}{S_n(f)}\right] \tag{3.19}$$

A signal is detected when, $\rho(t) \geq \rho_{thr}$. Where, ρ_{thr} is the chosen threshold[22].
Gravitational Waves experiments like LIGO use Matched Filtering based algorithms such as 'FINDCHIRP'[22] to detect and analyze the GW signals.

3.2. Frequency domain

In this section, we will explore the signal detection methods operating in the frequency domain.

3.2.1. Baseline

In the frequency domain, the excess power method can be used to detect unknown transient signals in noisy data [23]. Unlike matched filtering, the excess power method does not require a template and only requires noise data. The formula for calculating excess power is give by

$$\int (Data - \langle \text{Noise} \rangle)\, df = \text{Residual} \tag{3.20}$$

where *Data* is the test input vector containing noise and a potential signal waveform. $\langle Noise \rangle$ is the spectral noise average of the selected calibration noise data segment. Similar to the Matched Filtering, the signal is detected when Residual $> \rho_{thr}$, where ρ_{thr} is the chosen threshold.

The excess power method works without any prior assumptions and can be used to find the signal irrespective of its shape [23].

This method is computationally lighter and cheaper to implement; it is also more suited to experiments like GravNet, where the signal shape is unknown.

We used this method as our baseline to compare the results from the ML detection pipeline.

3.3. Machine learning based GW signal detection

In recent years, the idea of using Machine Learning for GW signal detection has gained significant traction. GravNet has also designed a Machine learning detection approach using time-domain noise-only data from the SUPAX cavity at Mainz.

3.3.1. Time-domain CNN LSTM classifier

In this section, we will see the architecture of a time-domain-based neural network designed to detect GW signals.

As shown in the figure 3.1, the neural network consists of Convolution Neural Network (CNN) layers as well as Long-Short-Term-Memory (LSTM) layers.

First, the input vectors are passed through a CNN layer that extracts local features from the time series data. The second CNN layer is added to further integrate the extracted features and to produce a higher-level representation. The output of the second CNN layer is passed to the LSTM layer. LSTM models the temporal dependencies across the sequence. The LSTM layer is followed by two fully connected dense layers. The dense layers apply a non-linear transformation to the extracted temporal features, and the second layer outputs the final classification score.

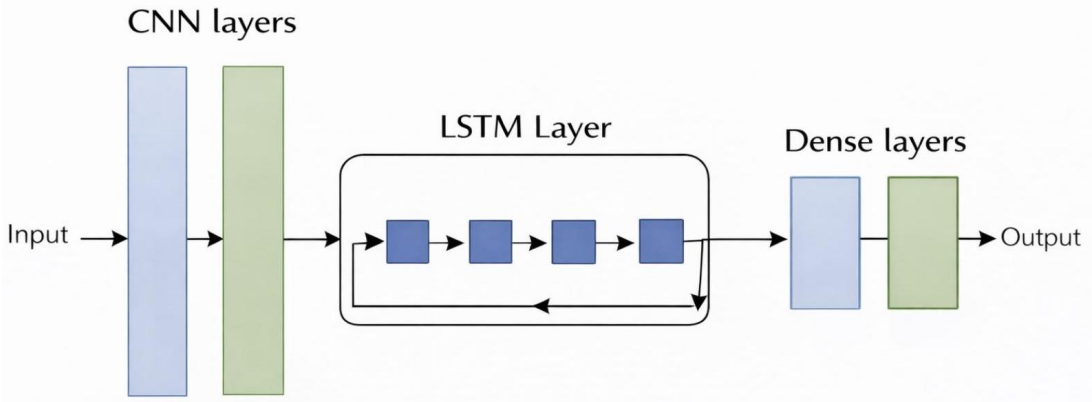


Figure 3.1.: Schematic diagram of CNN-LSTM classifier architecture

The architecture is designed to be a hybrid neural network as, in theory, it allows the model to learn the temporal information as well as to retain the strong feature extraction capability of CNNs[24].

To train the CNN LSTM classifier, the data is preprocessed as follows:
The noise data from the SUPAX cavity at Mainz is first divided into three segments, namely training, validation, and testing datasets. Within two data files of 8 seconds each, a total of 19,000 simulated signals are injected at random starting times: 15,000 signals in the training segment, and 2,000 signals each in the validation and testing segments. The signal simulation and signal injection process used here differs from the one described in this thesis; the details can be found in this bachelor's thesis.

The exact signal location is tracked by using a Boolean mask that flags all data points containing a simulated signal. Then each segment is further divided into an overlapping input vector of 1024 data points with a step size of 102. If an input vector contains at least one signal, it is labeled as signal containing; otherwise, it is labeled as noise only.

The model is trained as a binary classifier to distinguish between signal and noise

samples. In this setting, the standard choice is to use a sigmoid activation function in the final layer and binary cross-entropy as the loss function. But a different approach was taken here. As the sigmoid's output has saturation in the tails, it reduces the numerical resolution. It makes achieving extremely low false positive rates ($\leq 1/\text{yr}$) difficult. For this reason, a linear activation function is used in the final dense layer, directly outputting raw logits. In addition, the binary focal cross-entropy function is used instead of the binary cross-entropy as a loss function. This approach proved effective in suppressing large noise outliers by placing relative emphasis on them through a weight balancing factor of $\alpha = 0.25$ and a focal factor of $\gamma = 4$, where the latter increases the importance of hard-to-detect examples during training.

During the training of the CNN LSTM classifier, the simulated signals corresponding to PBH mass of $10^{-8}M_{\odot}$ and pure noise samples are used. More details on the topic can be found in this research report and this bachelor's thesis.

3.3.2. Frequency domain

Frequency-domain neural networks offer several advantages over their time-domain counterparts, most notably the presence of more consistent noise patterns and average noise floor. In the frequency domain, the AE can more effectively learn these patterns and reconstruct the noise structure, leading to a significant improvement in detection efficiency. Therefore, in this thesis, we designed and trained frequency-domain neural networks for signal detection.

In this section we will explore the architecture and training of CNN autoencoder as well as multi-view classifier.

CNN Autoencoder

The terms CNN autoencoder and AE are used interchangeably in this thesis to refer to this model. The architecture of the CNN autoencoder used in this thesis is as follows:
Architecture
The model consists of a 1D Convolution Autoencoder for anomaly detection. It is designed for unsupervised anomaly detection on 1-dimensional spectral data with an input length of 78. The autoencoder consists of two parts: an encoder and a decoder.

The encoder consists of five 1D convolutional layers with a kernel size of 3 and ReLU activation[25]. These layers use 1, 8, 16, 8, and 4 filters, respectively. The first convolutional layer uses valid padding, reducing the input length from 78 to 76. The following convolutional layers use the same padding.

Each convolutional layer is followed by a max-pooling layer with a pool size of 2, which progressively reduces the dimension while retaining the most relevant features. After the final pooling operation, the feature maps are flattened and passed through a dense layer with a single neuron. This produces a one-dimensional latent representation, which serves as the bottleneck of the autoencoder. Thus, the dimensional flow of the encoder is

$78 \to 76 \to 38 \to 19 \to 10 \to 5 \to 3 \to 1$.

The decoder is the reverse of the encoder; it reconstructs the signal from this latent variable. It starts with a dense layer that expands the latent representation into 16 units, which are then reshaped into a 4×4 feature map. This is followed by a sequence of 1D convolutional layers with filters of 4, 8, 16, 8, 8, and 1. ReLU activation is used in all hidden layers, and sigmoid activation is used in the final output layer. Between each convolutional layer, upsampling by a factor of 2 is applied to increase the feature-map length and gradually restore the original input dimension. The decoder's dimensional flow is $1 \to 16 \to 4 \to 8 \to 16 \to 32 \to 64 \to 128 \to 78$. The full architecture of the AE can be seen in Figure 3.2.

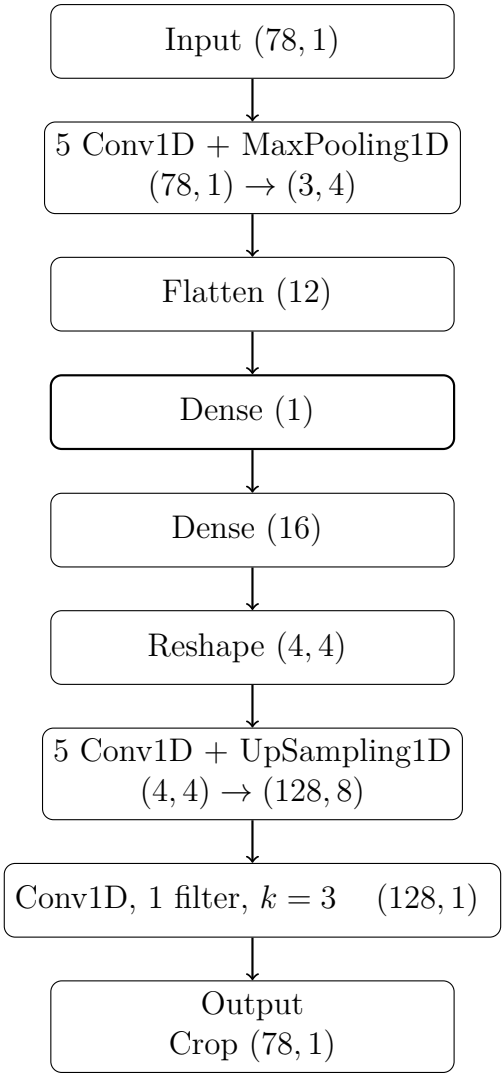


Figure 3.2.: Schematic diagram representing the architecture of 1D CNN Autoencoder designed for anomaly detection.

From figure 3.2, we can see that the encoder reduces the dimensions of the input vector from 78 $\rightarrow$ 1 and the decoder upsamples it from 1 $\rightarrow$ 128. The decoder is not the exact opposite of the encoder, and the upsampling process produces an output dimension of 128. Thus, the final layer has to be cropped to the first 78 elements, ensuring that the output has the same length as the input.

After a careful review of the architecture, future modifications to the decoder may be considered. In particular, the decoder could be redesigned to more closely mirror the encoder, producing an output of length 78 directly rather than relying on a final cropping operation. Such a modification may improve reconstruction consistency and could potentially enhance the signal detection efficiency, although this would need to be verified through further testing.

Training

The AE is trained on noise data obtained from the SUPAX cavity at Mainz. The noise data is in IQ format and saved in '.tiq' files. The time series data is first divided into segments of 1024 data points, and the Fast Fourier Transform (FFT) is performed on each segment individually to convert them into the frequency domain.

Finally, only the spectral data within the FWHM of the cavity is selected and used for training the AE. The resulting frequency domain input vectors are 78 data points long. Figure 3.3 shows an example of the frequency domain noise data and the FWHM of the cavity.

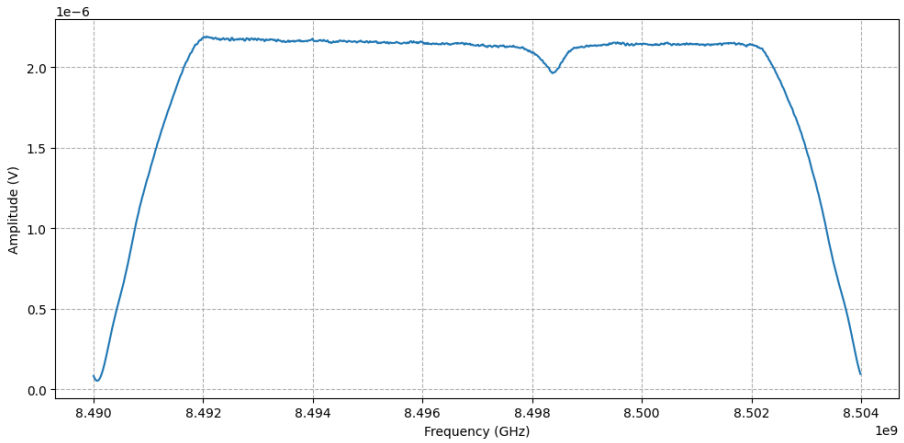


Figure 3.3.: Frequency domain noise data with cavity resonance shape.

These input vectors are divided into training and validation data sets in the ratio of 80:20. The minimum and maximum values of the training data are found by using the function '$tf.reduce_min$' to normalize all of our data to the interval of [0,1]. For our dataset, the resulting minimum and maximum values are 3.174×10^{-14} and 2.67×10^{-5}, respectively.

This step is crucial to avoid instabilities during training and to ensure that differences in the orders of magnitude across input vectors do not introduce any biases.

Additionally, we calculated the mean of every 1000 input vectors in our training dataset to use it as the output of our AE while training. This step ensures that the AE primarily learns the average of our input data and reconstructs the average shape and distribution of noise. Since noise is random, our goal with reconstruction is to learn the average shape of the distribution. With this, our pre-processing is complete. Now the AE is trained with input vectors from the training dataset and the above mentioned average of 1000 vectors as outputs.

In addition, the further testing is performed by normalizing the test data with the same minimum and maximum values mentioned above to ensure consistency in the orders of magnitude of data.

Multi-view Classifier

The architecture of multi-view classifier used in this thesis is as follows:
Architecture In this thesis, a binary classifier with multi-view feature inputs is used to classify vectors as either signal-containing or pure noise. The classifier has 3 branches, which receive 3 different inputs for each sample. The first branch receives frequency-domain representations of two classes: pure-noise samples and noise samples with an embedded signal. The dataset is class-balanced, with each class contributing 50% of the samples. The second branch receives AE reconstructions corresponding to the input vectors provided to the first branch. Lastly, the third branch receives the reconstruction loss computed from the first-branch input vector and its corresponding AE reconstruction. Here, the reconstruction loss values are calculated by using the sum loss function given in the equation 3.25. This particular loss function was chosen as it performs well in AE's detection efficiency test results; more details and explanation about this can be found in section 3.3.3.

All three branches are passed through separate deep neural network layers. Each branch maps the corresponding feature into a branch specific latent representation. These latent representations are then combined into a concentrated feature vector, which serves as the input of the final classification head.

The first branch receives 78-dimensional input vectors containing pure noise data and noise with an embedded signal, in equal parts. The data is passed through 3 fully connected layers with 64, 32, and 16 neurons, with ReLU activation. Thus, the original input vector is mapped into a compact 16-dimensional representation.

The second branch receives the corresponding AE reconstruction, which is also represented as a 78-dimensional vector. This branch has the same fully connected structure as the first branch and maps the reconstructed input to a 16-dimensional latent representation.

The third branch receives the corresponding reconstruction loss value with a shape of (batch size,1), where batch is the number of samples processed simultaneously. The branch consists of 2 dense layers of 16 and 8 neurons with ReLU activation function. The final output of this branch is an 8-dimensional representation.

Once the outputs from all 3 branches are obtained, they are combined in a concentration

layer. This vector has 16+16+8= 40 dimensions.

The combined representation is passed through a classification head. It consists of 3 dense layers. First with 64 neurons, followed by a dense layer of 32 neurons. Then we have a small dropout layer with a rate of 0.1 to reduce overfitting. Then the final dense layer has only one neuron and a sigmoid activation function. The resulting output is one-dimensional and represents the probability of the positive class, with values ranging from 0 to 1. The schematic diagram of this binary multi-view classifier is given in the figure 3.4.

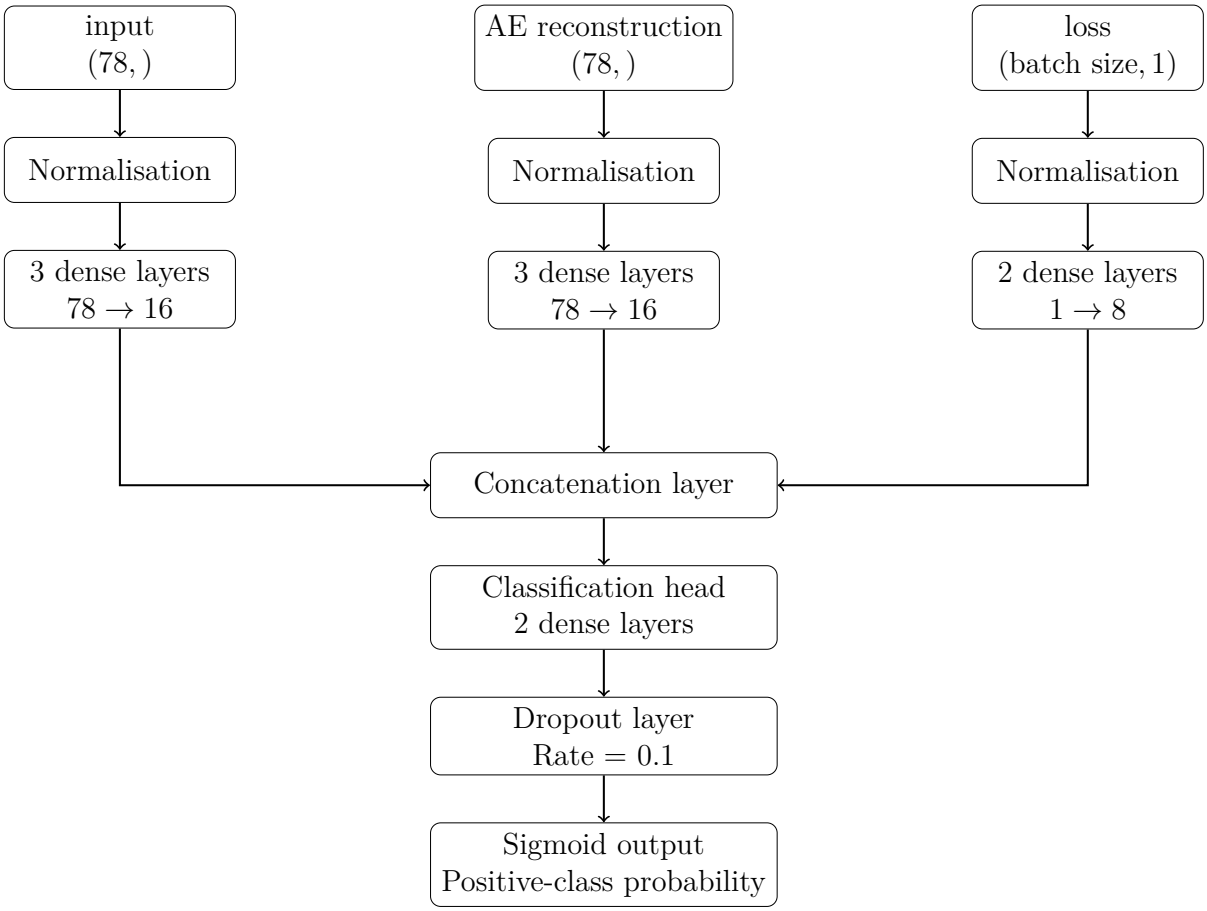


Figure 3.4.: Schematic diagram of the binary multi-view classifier architecture.

Since the labels are known from the injection procedure, the classifier is trained to distinguish between pure-noise samples and signal-containing samples in a supervised manner. The details of this training are as follows:

Training

To train the classifier, the noise data from the Supax cavity at Mainz was used. 50% of the data was used as it is, while the remaining half had a signal embedded in it. The pure-noise samples were labeled $y = 0$, and samples containing an injected signal were

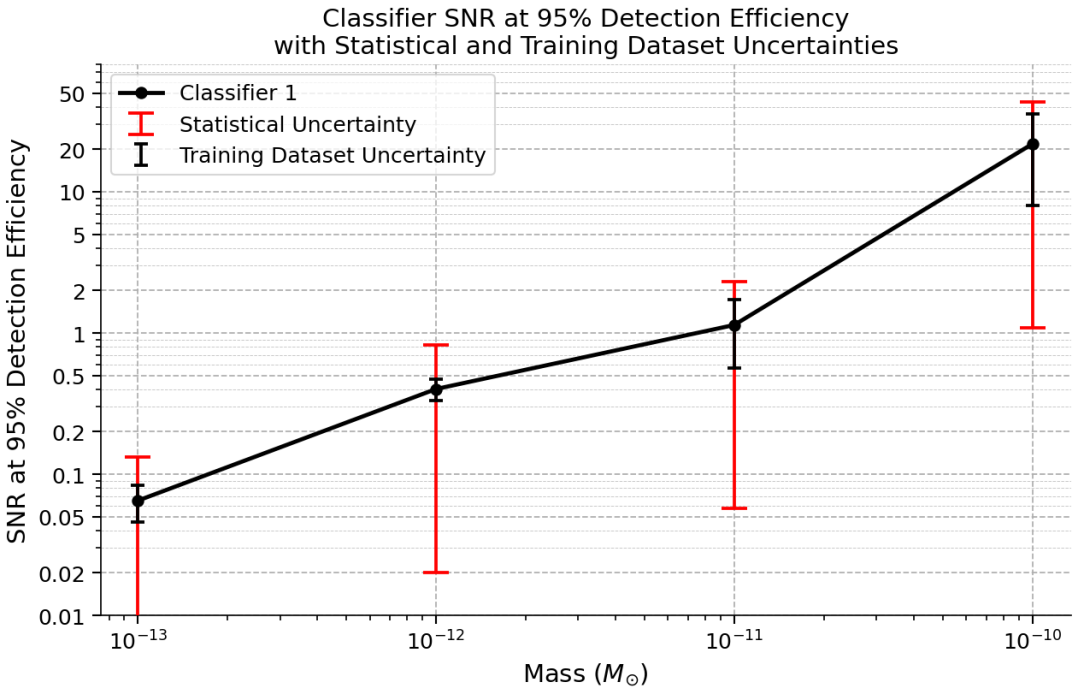


Figure 3.5.: Classifier 1 is the detection efficiency of the classifier trained with signals of SNR 1. Red error bars display the statistical uncertainty from 10 different trainings using the same training data of SNR 1. While the black error bars show the uncertainty arising from different training datasets with SNR of 1, 3,5 and 1 plus 5 in equal proportion.

assigned the label $y = 1$. This combined input will be represented by x_i^{input} Corresponding reconstructions (x_i^{rec}) of these input vectors were calculated with AE. Finally, the reconstruction loss values (x_i^{loss}) for each input vector and its corresponding reconstruction are computed by using the sum loss function. Thus, each training sample is represented as

$$\left(x_i^{\text{input}}, x_i^{\text{rec}}, x_i^{\text{loss}}, y_i\right), \tag{3.21}$$

where y_i represents the corresponding label.

During training, the classifier receives 3 input views, $(x_i^{\text{input}}, x_i^{\text{rec}}, x_i^{\text{MSE}})$, and it outputs a probability $\hat{y}_i$ that the sample belongs to the $y = 1$ class. The prediction is compared with the label value y_i given to the input vectors, by using binary cross-entropy loss. The model parameters are then updated by backpropagation using the Adam optimizer. The signals used to train the classifier correspond to the PBH mergers of mass, $10^{-13}M_\odot$, $10^{-12}M_\odot$, $10^{-11}M_\odot$, and $10^{-10}M_\odot$. Each signal has an SNR of 1. It should be noted that changing the SNR of training data has very little impact on detection efficiency.

To support the argument, the results in Figure 3.5 can be analyzed. The classifier was trained on signals with SNR of 1, 3, 5, as well as a mixed dataset with SNR of 1 and 5,

with 50% signals corresponding to each SNR. These signals make up half of the training dataset, and the rest is the noise data that remains unchanged. After each training, the classifier was tested for the SNR required to achieve 95% detection efficiency. Then the difference between the maximum and minimum SNR obtained during testing for each data point was noted. The difference in maximum and minimum SNR is the uncertainty arising due to the different datasets used for training and is shown in black error bars in figure 3.5. The classifier was then trained 10 times by using signals with an SNR of 1. The architecture, training dataset, or any other parameters of the classifier were unchanged during the process. The fluctuation in SNR required to achieve 95% detection efficiency was noted. The RMS of these SNR values is the statistical uncertainty, shown with red error bars in Figure 3.5.

It can be observed that the total range of statistical variations is much greater than the fluctuations caused by using various training datasets. Thus, the SNR of the training signals does not significantly affect the detection efficiency of the classifier. Consequently, the simplest training approach is adopted, in which signals with an SNR of 1 are used as the training dataset.

3.3.3. False Positive Rate

To test our detection pipelines, we first have to determine a threshold on our reconstruction loss and classifier's output. To obtain a suitable threshold, an appropriate False Positive Rate (FPR) must be established first.

FPR is the fraction of negative samples incorrectly labeled as positive by the detection pipeline. Mathematically, FPR can be written as

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}, \tag{3.22}$$

where FP stands for false positives, and TN stands for true negative samples.

A high false-positive rate can trigger too many false alarms. On the other hand, a low false-positive rate can make the threshold too high, leading to real signals being missed. It is therefore important to maintain a good balance between keeping false alarms low and maintaining high efficiency for true positive signal detection. Hence, an optimal FPR must be chosen.

In this thesis, an FPR of 1/year will be used for threshold determination and further analysis. The FPR is chosen arbitrarily based on the fact that the GravNet would collect data for roughly a year per run. It would be ideal to have no more than 1 false alarm during this period.

To determine the threshold, we first plot the probability density of reconstruction losses corresponding to the noise-only input vectors. A function is then fitted to the probability density distribution, and the threshold is determined such that the integral of the fitted function above the threshold equals the target false-positive probability for one year.

Mathematically, the expression is written as

$$\int_{thr}^{\infty} f(x)\,dx = FP_{prob}, \tag{3.23}$$

where thr is the threshold and FPR_{prob} is the false positive probability per year. Equation 3.23 defines the threshold as the point above which the expected fraction of background events matches the desired false-positive rate.

As each input sample is 1024 data points long, with a duration of 7.31μs, in 365 days, 4.314 $\times 10^{11}$ input samples would be observed. So, the false positive probability (FP_{prob}) is calculated as

$$\frac{1}{4.31 \times 10^{11}} = 2.319 \times 10^{-12}$$

.

The fit function chosen for the threshold calculation must provide a satisfactory description of the given probability density distribution with a reduced χ^2 of 1 or very close to it. If a suitable function is not found to describe the full distribution of the data, a tail fit is performed instead. For example, in figure 3.6, the Gaussian fit for the full distribution has a reduced χ^2 of 354.084, significantly far away from the ideal value of 1.

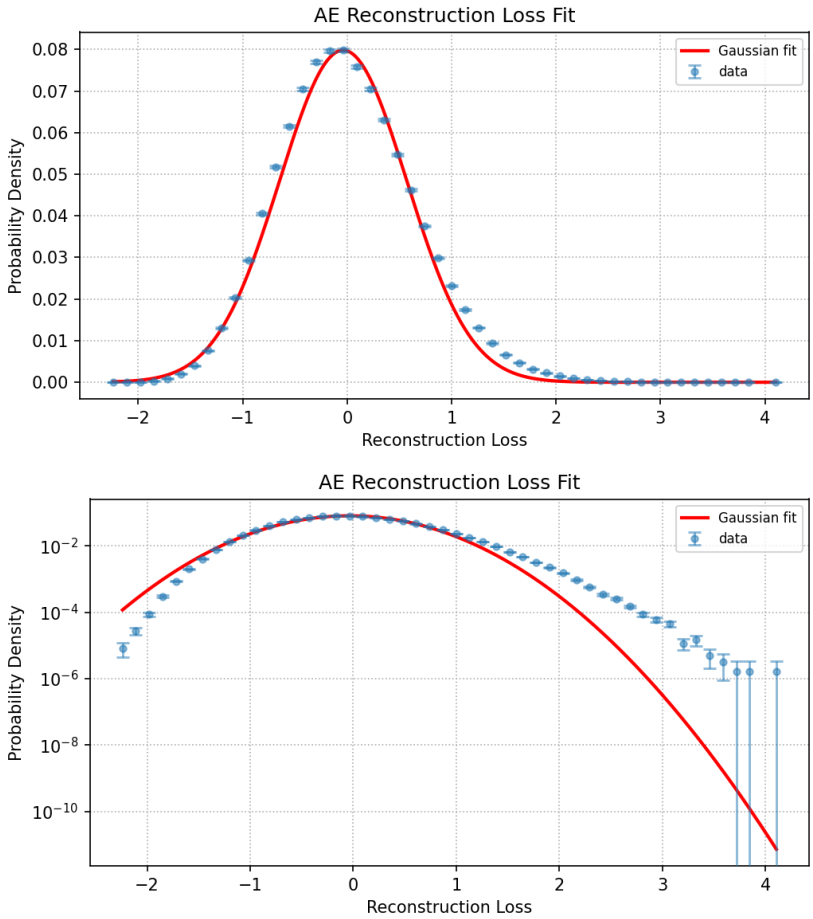


Figure 3.6.: In this figure, AE reconstruction loss's probability density is shown in arbitrary units. In the upper image, the data is plotted on a linear scale, and in the bottom image, the same data is shown on log-log axes. A Gaussian fit is attempted on the data; however, it does not describe the distribution well.

As the noise data used in the training process has white noise characteristics originating from various electronic equipment used in the readout, the full probability density distribution was expected to be a Gaussian. However, figure 3.6 shows that the Gaussian fit does not adequately describe the characteristic behavior of the data. This is because the cavity resonance shape is present within the FWHM of the noise file. It alters the noise behavior, resulting in a deviation from pure white noise. Hence, we proceed with a tail fit by using an exponential function for the reconstruction loss data. The exponential fit can be described by

$$y = A \exp^{\beta} . \tag{3.24}$$

The fit and its parameters are shown in the figure 3.7. From the probability distribution of training data, the threshold for AE (Sum loss) is 7.282 ± 0.734.

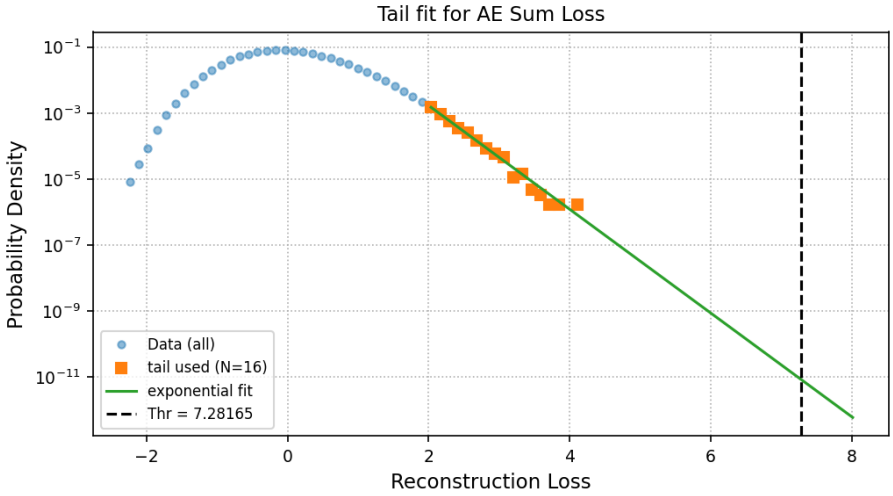


Figure 3.7.: Exponential fit to the tail of the probability density of the AE reconstruction loss, where the reconstruction loss is computed using the sum-loss function. The fit parameters of the function are: $A = 2.361 \pm 4.097$, $\beta = -3.62 \pm 0.077$ and reduced χ^2 of 1.023.

The probability distribution in figure 3.7 is calculated by using the sum loss function, which is given by the formula,

$$\text{Sum Loss} = \sum_{j=1}^{N} (x_{ij} - \hat{x}_{ij}) \,, \tag{3.25}$$

where N is the number of data points and x_{ij} denotes the j-th input vector, while $\hat{x}_{ij}$ denotes the corresponding reconstruction.

The distribution of the reconstruction loss's probability density is based on the loss function used to compute it. Therefore, changing the loss function results in different distributions and, consequently, different threshold values. To obtain the best results, several loss functions were tested. The detailed loss functions and their associated thresholds are tabulated in Table 3.1.

Table 3.1.: Various loss functions and their corresponding thresholds

Loss Function	Formula	Threshold
Sum Loss	$\sum_{j=1}^{N} (x_{ij} - \hat{x}_{ij})$	7.281 ± 0.734
Mean Squared Error (MSE) Loss	$\frac{1}{N} \sum_{j=1}^{N} (\mathrm{x}_{ij} - \hat{\mathrm{x}}_{ij})^2$	0.052 ± 0.031
Mean Absolute Error (MAE) Loss	$\frac{1}{N} \sum_{j=1}^{N} \lvert\mathrm{x}_{ij} - \hat{\mathrm{x}}_{ij}\rvert$	0.096 ± 0.009
Max Loss	$\max_{1 \leq j \leq N} (\mathrm{x}_{ij} - \hat{\mathrm{x}}_{ij})$	1.299 ± 0.127

The uncertainty in the threshold is estimated from the two dominant sources of systematic and statistical uncertainty. The systematic uncertainty arises from the error propagation

of the fitted parameters. While the statistical uncertainty is the difference between the various fit functions used to describe the same loss distribution. Statistical uncertainty makes a more significant contribution in most cases in this thesis. Since the two sources of uncertainty are independent of each other, we take the quadrature sum of these two components to compute the total uncertainty. The total uncertainty can be expressed as

$$\sigma_{\text{tot}} = \sqrt{\sigma_{\text{stat}}^2 + \sigma_{\text{syst}}^2}, \tag{3.26}$$

where σ_{tot} is the total uncertainty, σ_{stat} denotes the statistical uncertainty from the error propagation of the fit parameters, and σ_{syst} represents the systematic uncertainty estimated from the variation between different fit functions. From 3.1 it can be observed that, each loss has a different threshold, reflecting the distinct mathematical properties of the corresponding loss formulation. Since the loss functions quantify the reconstruction loss in different ways, their sensitivity to anomalous patterns also differs.

MSE is the most standard loss function used in anomaly detection, but it can be less effective when the anomaly is distributed across multiple frequency bins in the input vector rather than concentrated in a few large deviations. In this case, the sum loss is the more suitable for the purpose of the anomaly detection. Also, the uncertainty in the sum loss's threshold is smaller than the MSE loss in percentage, making it more reliable. Thu, sum loss function will be used throughout this thesis along with the threshold noted in Table 3.1.

Similarly, the threshold for the excess power method is calculated by using the distribution of residuals, and the resulting threshold is 2.144 ± 0.452. The fit parameters for the tail fit can be found in figure 3.8.

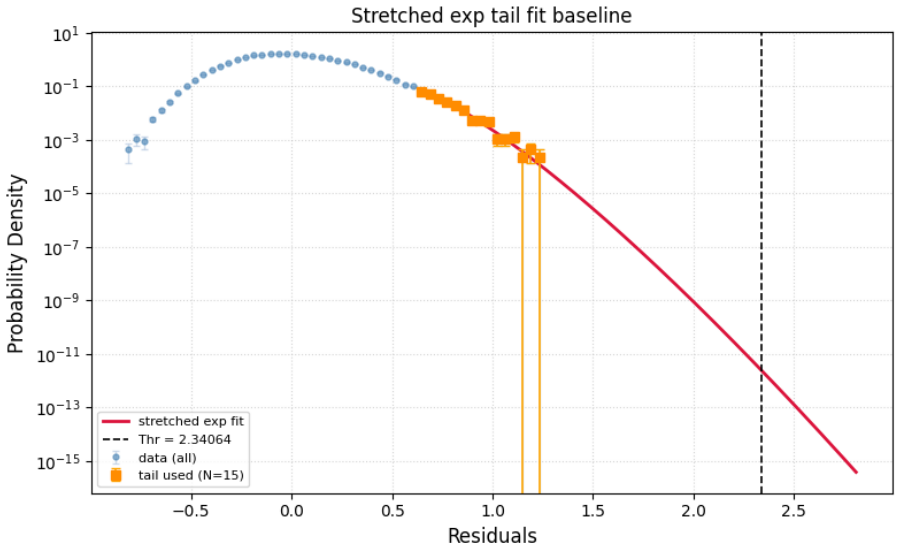


Figure 3.8.: Tail fit on baseline's residuals with stretched exponential function. The fit parameters are: A= 6.085 $\times 10^{-2}$, $\tau = tau = 0.138 \pm 0.008$, $\beta = 1.269 \pm 0.079$ and reduced $\chi^2 = 1.2910$

Here, A= 6.085 $\times 10^{-2}$, $\tau = tau = 0.138 \pm 0.008$, $\beta = 1.269 \pm 0.079$ and the reduced χ^2 is 1.2910 .

For the matched-filtering technique, the first step was to calculate the peak value for each input vector. The peak value is defined as the largest magnitude of the matched-filter output corresponding to that input vector. The threshold for the matched filtering technique is determined by the distribution of the probability density of the peak values. Unlike AE, for matched filtering, the threshold changes for each template. In this case, only one template corresponding to the signal waveform of the $10^{-11} M_{\odot}$ was used. To obtain the threshold for this template, we first plotted the probability density distribution of matched-filter peak values shown in Figure 3.9.

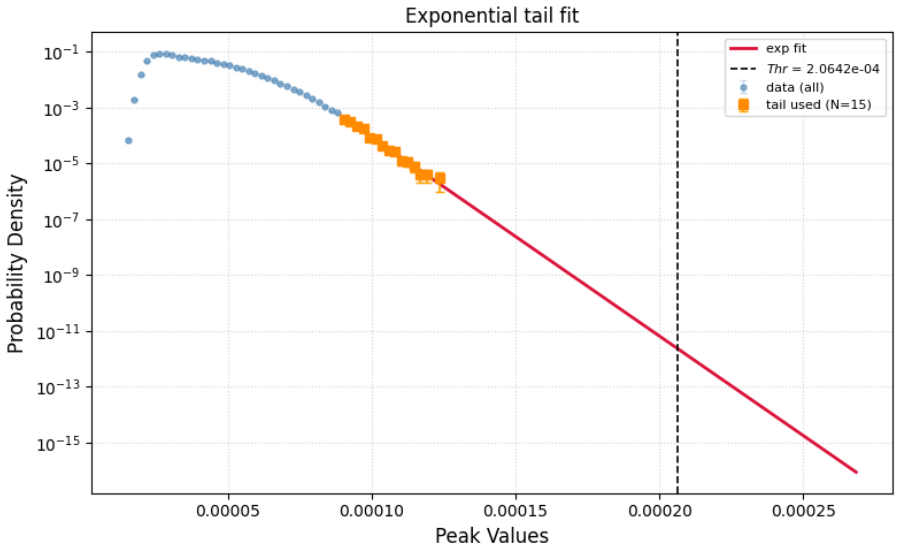


Figure 3.9.: Exponential tail fit on the probability density of the peak values. The fit parameters are: A= $1.224 \times 10^3 \pm 6.005 \times 10^2$ $\beta = 6.089 \times 10^{-6} \pm 1.901 \times 10^{-7}$

The threshold was calculated by using Equation 3.23 and for FPR of 1/yr is $2.064 \times 10^{-4} \pm 2.923 \times 10^{-5}$.

3.3.4. FPR for classifier

Ideally, the FPR for the classifier should be 1/yr, similar to the AE. However, in practice, achieving such a low FPR with limited data is not possible for the classifier used in this thesis.

To estimate the threshold corresponding to a target FPR per sample lower than the classifier's observed false positives rate, we first identify all true negative samples and plot their probability density as a function of the classifier output. Since the final layer of the classifier uses sigmoid activation, the output is limited to the range (0,1). Due to this limited output range, it is extremely difficult to extrapolate the probability density distribution of classifier output and find an FPR that is far below the currently observed number of false positives.

The probability distribution of the observed true negatives' outputs is shown in the figure 3.10. It can be seen that the probability distribution extends to the classifier output of 0.989, very close to 1, leaving very little higher output region for a reliable tail fit or extrapolation. Therefore, the threshold cannot be set based on the FPR of 1/yr, the criterion used for the AE.

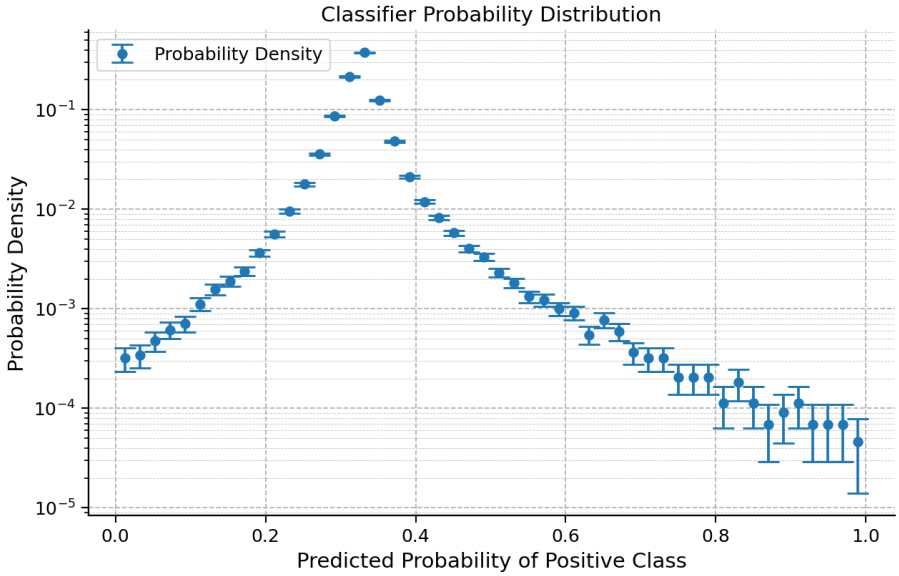


Figure 3.10.: Probability density as a function of the classifier output for true negative samples from training dataset. The distribution stretches till the output probability of 0.989, very close to 1. Thus, the ability to extrapolate the data or fit a tail function is very limited.

Instead, we use the classifier's output to determine the FPR and threshold corresponding to it. From Equation 2.27, we can find the estimated probability of the positive class and set a decision threshold on it. Samples with an output score above this threshold are classified as signal containing samples, while samples below the threshold are classified as pure noise samples. The threshold can be determined from the Receiver Operating Characteristic (ROC) curve.

The ROC curve is obtained by varying the decision threshold over the classifier output range and evaluating the classifier performance at each threshold. For every threshold value, the predicted labels are compared with the true labels in the validation set. This gives the number of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). From these quantities, the True Positive Rate (TPR) and the false positive rate are calculated as TPR = TP / (TP + FN) and FPR = FP / (FP + TN), respectively. The ROC curve then plots the TPR against the FPR for all considered threshold values. Figure 3.11 shows the ROC curve corresponding to the classifier used in this thesis.

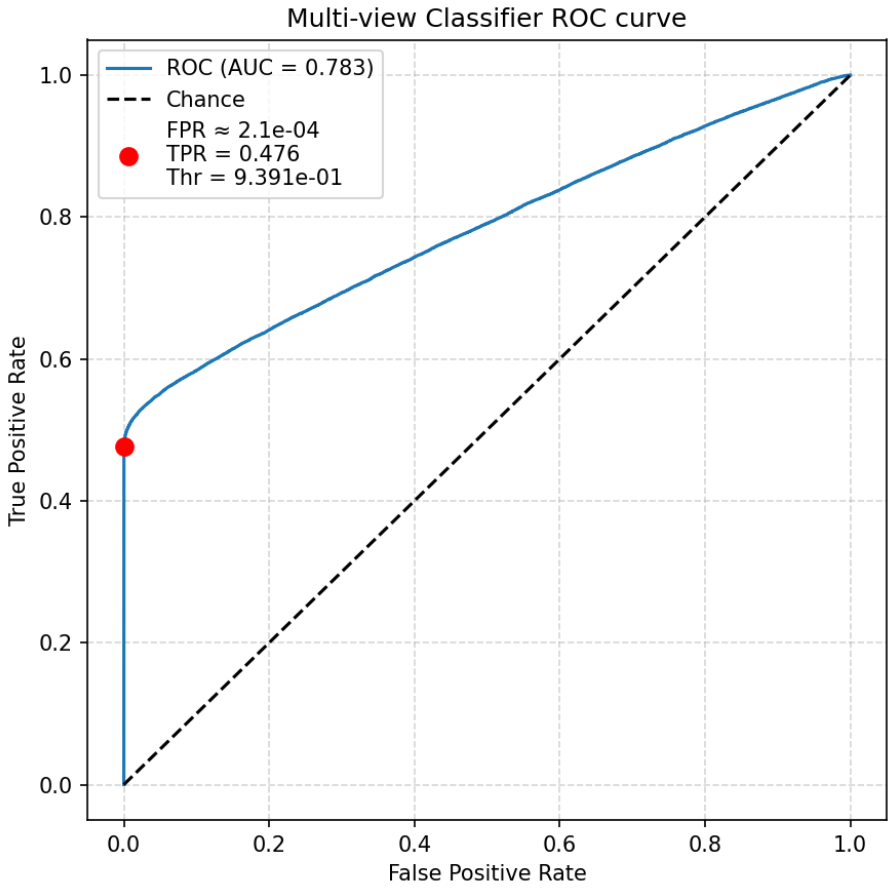


Figure 3.11.: ROC curve for the classifier with AUC of 0.783. The desired FPR is 2.057×10^{-4}, and the corresponding TPR is 0.476. The resulting threshold on classifier output is $0.939^{+0.000084}_{-0.011992}$.

By taking into account all practical constraints, an FPR of 2.057×10^{-4} per sample was chosen for this thesis, which corresponds to the FPR of 8.58 $\times10^{7}$/ yr. The choice was made as it is well within the observed false positive range per sample and has a fairly high TPR while maintaining a low FPR. The resulting threshold is $0.939^{+0.000084}_{-0.011992}$

To estimate the uncertainty in the obtained threshold, the FPR was varied by one step above and below its value on the ROC curve, and the corresponding changes in the threshold were taken as the upper and lower uncertainties.

The AUC-ROC value of 0.783 indicates that the classifier has moderately good discriminative ability. A signal has 78.3% probability of being assigned a $y = 1$ class by the model, which is better than a chance. This is not an ideal ROC curve for a classifier, and there is scope to improve the results in the future.

It should be noted that, for a supervised classifier with a sigmoid output, it will always be challenging to extrapolate the probability distribution of true negatives to obtain an extremely low FPR with a few hours of calibration data. Thus, for a classifier, we always

have to choose a higher FPR, which is not an issue as long as the further analysis of flagged input vectors is manageable.

4. Results

In this section, we will analyze the detection efficiencies of ML-based detection pipelines and compare the results with baseline and matched filtering methods. Moreover, we will also test the detection methods against noise fluctuations in the frequency domain.

4.1. Detection efficiency

In this section, we will analyze the detection efficiency of the Autoencoder discussed in section 3.3.2. The corresponding results will also be compared with the baseline and time domain based detection pipelines discussed in sections 3.2.1 and 3.3.1, respectively.

To keep the comparison fair, the FPR of 1/ yr was used for all detection pipelines. The input vector size is 1024 data points, and for the baseline and AE, the data is further trimmed in the frequency domain to include only the FWHM of the cavity (78 data points). For testing, 109375 signals were used per component mass. The exception to this case are the signals corresponding to the PBH mass of $10^{-13}M_{\odot}$ with 211 test signals, $3.162 \times 10^{-13}M_{\odot}$ with 1434 signals, and $10^{-12}M_{\odot}$ with 9775 signals.

For the CNN LSTM classifier, the input vector size remains the same, but overlapping input vectors were used with a step size of 102. Moreover, the simulated signals used to test the CNN-LSTM classifier are generated and injected into the noise data using a different procedure from the one described in this thesis. More details can be found in this bachelor's thesis and in this research project.

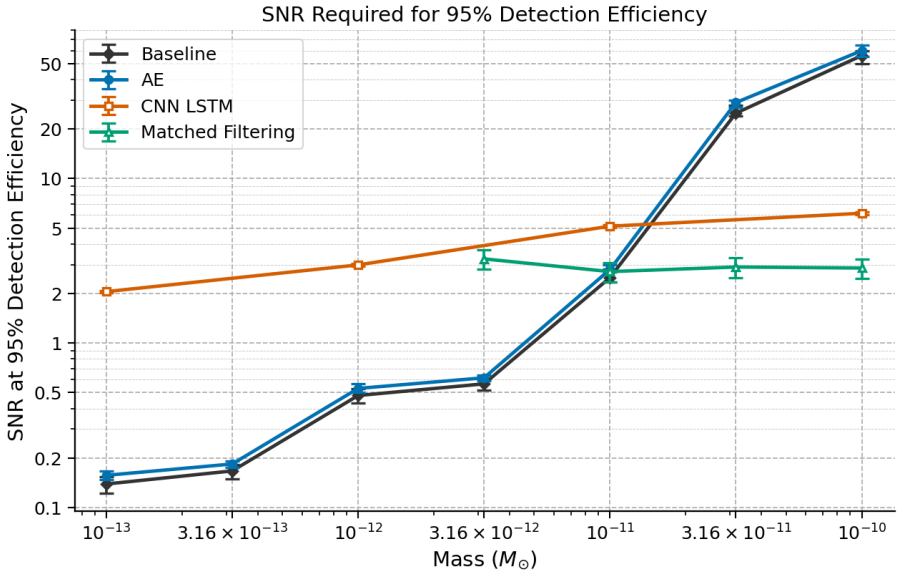


Figure 4.1.: PBH mass vs the SNR required to achieve 95% detection efficiency. Baseline achieves the 95% detection efficiency at the lowest SNR for signals corresponding to PBH masses in the range of $10^{-13}M_\odot$ to $10^{-11}M_\odot$. AE requires a slightly higher SNR, but the performance is within the uncertainty range. The time domain methods require much higher SNR than the frequency domain methods in the same mass range. But for higher mass PBH merger signals, Matched filtering performs better than AE and baseline. At all tested data points, it requires an SNR between 2 and 3 to achieve 95% detection efficiency. AT all points, the CNN LSTM classifier requires a higher SNR than matched filtering to achieve 95% detection efficiency. Thought it performs better than AE and baseline at the mass point of $10^{-10}M_\odot$ and $3.162 \times 10^{-11}M_\odot$.

Figure 4.1 shows the PBH mass vs the SNR required to reach 95% detection efficiency. Among the three tested methods, the baseline method reaches 95% detection efficiency at the lowest SNR for component masses between $10^{-13}M_\odot$ to $3.162 \times 10^{-11}M_\odot$. The Autoencoder with sum loss performs similarly to the baseline, but requires slightly higher SNR to reach 95% detection efficiency. However, the error bars of the AE and baseline methods overlap, indicating that their performances are comparable within the uncertainty.

The error bars were estimated by varying the corresponding thresholds within their uncertainty range and repeating the measurements. The resulting variation in the extracted reading was taken as the uncertainty on that data point for AE, baseline, and matched filtering methods.

The matched-filtering method was used to detect signals with component masses ranging from $3.162 \times 10^{-12}M_\odot$ to $10^{-10}M_\odot$. Signals with masses below this range have durations longer than the length of a single input vector. As a result, generating a template and sliding it across multiple input vectors would be computationally expensive, and it was

not feasible within the resource limitations of this thesis.

All matched-filtering results were obtained using a single template corresponding to the PBH mass of $10^{-11}M_{\odot}$. As discussed in the section 3.1.1, the method used time domain data.

For signals corresponding to $10^{-11}M_{\odot}$, matched filtering requires SNR similar to AE and the baseline method to achieve 95% detection efficiency.

At the mass point of $3.162 \times 10^{-12}M_{\odot}$, the SNR required for matched filtering to achieve 95% detection efficiency remains approximately unchanged. In contrast, the SNR required by AE and the baseline method decreases significantly at this mass point. For higher masses, specifically $3.162 \times 10^{-11}M_{\odot}$ and $10^{-10}M_{\odot}$, the SNR required by matched filtering at 95% detection efficiency remains between 2 and 3, whereas the corresponding SNR values for AE and baseline are substantially higher and above 50.

For a PBH mass of $10^{-11}M_{\odot}$, matched filtering uses the optimal filter with the exact match of the template. Thus, this is the best result we can obtain by using the matched filtering method, and it is comparable with the AE's performance. It should be noted that for all tested signals, the SNR required to achieve 95% efficiency shows minimal change, and this behavior is similar to the CNN LSTM classifier used in the time domain. However, only one template was used to detect all signals corresponding to the range of $3.162 \times 10^{-12}M_{\odot}$ to $10^{-10}M_{\odot}$; they all belong to the inspiral phase of the PBH merger and do not have merger or ringdown information in the simulated waveform. This creates a unique situation in which one template can still match different simulated signals. Another factor contributing to it is the multiplication of noisy data with the complex conjugate of the signal template in the frequency domain. Due to this step, the phase information is lost, and one template is able to extract signals corresponding to different PBH masses.

In a practical scenario, a signal of completely unknown shape might be missed by the matched filtering detection pipeline.

The time domain CNN LSTM classifier achieves the 95% efficiency at highest SNR for PBH mass range of $10^{-13}M_{\odot}$ to $10^{-11}M_{\odot}$. $10^{-10}M_{\odot}$ is the only data point where the CNN LSTM classifier reaches the 95% detection efficiency at a lower SNR than AE and baseline. At the same data points, the AE and baseline methods show a sudden increase in SNR corresponding to the 95% detection efficiency. This might be due to the fact that, during signal simulation, the resonance filter is individually applied to every input vector of 1024 data points. First, the signal is Fourier transformed to the frequency domain and then multiplied by the resonance response. This operation is mathematically equivalent to circular convolution, where the input vector is treated as periodic. Thus, when a signal is near the boundary of the input vectors, the resonant response can wrap around and reappear at the beginning of the same input vector rather than extending naturally into the next. The signals with a small time duration are affected the most, for example, a signal corresponding to the component mass of $10^{-11}M_{\odot}$ is approx. 6 data points long, while the signal corresponding to the PBH mass of $3.162 \times 10^{-11}M_{\odot}$ is approx. 28 data points long. These Signals have a much higher SNR at 95 % detection efficiency. On the other hand, longer signals are less sensitive to such an artifact, as only data points near

the boundary of the input vectors are affected, and most of the signal remains intact.

It should be noted that this effect arises purely due to the signal simulation algorithm; in practical cases, a resonance filter is not applied to each input vector individually, and it is an inherent property of the detector. Thus, there won't be any artifacts arising from signal wrapping.

As the signals used to test the CNN LSTM detection pipeline are not injected to each input vector individually and no resonance filter is applied on the finite input vector of 1024 data points, the signal wrapping artifact is not present in them.

Thus the AE and baseline detection pipelines might have a lower SNR at 95% detection efficiency than shown in the figure 4.1. A more realistic signal simulation can be used in future research to further study the detection efficiency of AE.

4.2. Classifier Results

In this section, we will analyze the results from the classifier. The details about its architecture and training can be found in the section 3.3.2. As discussed in section 3.3.4, an FPR of 8.869×10^{10}/ yr was used to obtain the results from the classifier. The FPR used in this section is significantly higher than the FPR of 1/yr used in the rest of this thesis. In order to fairly compare classifier results with other detection pipelines, we also increased the FPR for the AE and baseline detection methods to 8.869×10^{10}/ yr. Due to the limitation of noise data files, the classifier is trained and tested with simulated signals corresponding to the PBH masses of $10^{-10}M_{\odot}$, $10^{-11}M_{\odot}$, $10^{-12}M_{\odot}$ and $10^{-13}M_{\odot}$.

The PBH mass vs 95 % efficiency results are shown in the figure 4.2.

It can be observed that the classifier performs very similarly to the baseline and AE in the component mass range of $10^{-13}M_{\odot}$ to $10^{-11}M_{\odot}$.

The uncertainties of the classifier are calculated from the systematic uncertainty coming from threshold and from the statistical uncertainty which is calculated as

$$\sigma_{\text{stat}} = \frac{y}{\sqrt{N_{\text{eff}}}}, \tag{4.1}$$

where y is the SNr at 95% detection effiency and N_{eff} is the no of signals used for testing. The quadrature sum of these two is the final uncertainty in classifier's results. The uncertainty in AE and baseline's results are calculated similarl to the section 4.1.

The PBH mass of $10^{-10}M_{\odot}$ classifier shows some improvement over AE and baseline. As discussed in the section 4.1, these particular signals have a signal wrapping artifact that is causing an extremely high SNR to reach 95% detection efficiency. As the classifier is a supervised neural network, it might be better at differentiating signal from noise even when artifacts are present. This is an advantage that might be significantly beneficial for practical signal detection.

Though the classifier shows promising results, due to the limitation of data in this thesis, a proper conclusion about the classifier's abilities cannot be drawn. Further research is

needed in order to explore the capabilities of a classifier for HFGW signal detection.

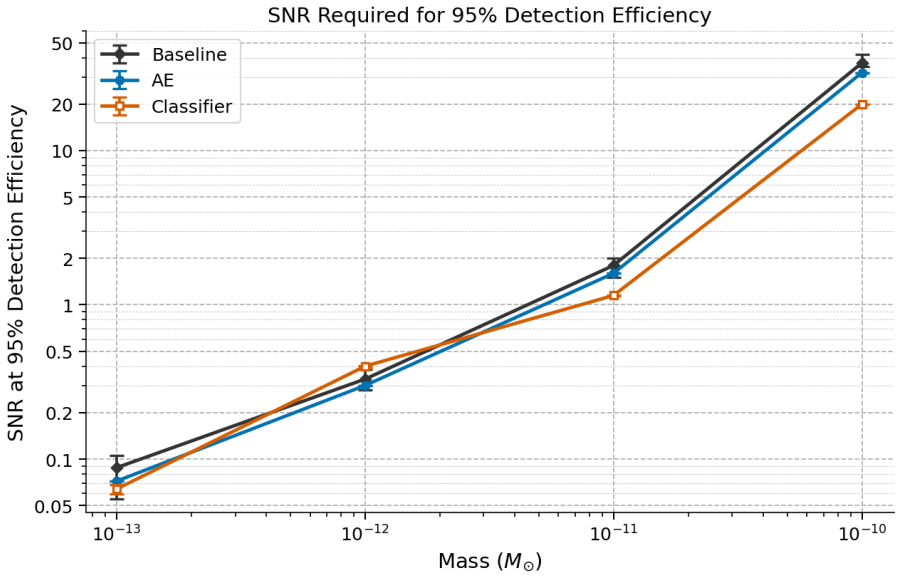


Figure 4.2.: PBH mass vs SNR required to achieve 95% detection efficiency. FPR of $8.869\times10^{1}0/$ yr was used to obtain the results for all three detection pipelines.

4.3. Offset

In a practical scenario, the noise data from an RF cavity varies over time, creating an offset in the frequency domain. Figure 4.3 shows the fluctuation of the noise over an observation period of 6 hours. The observed offset is from a minimum of 0.03% to a maximum of 0.3% of the noise floor.

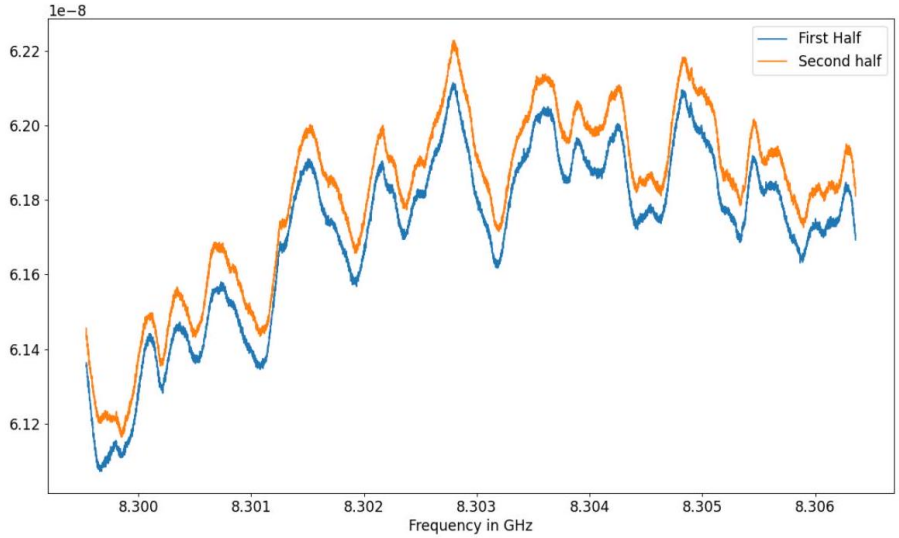


Figure 4.3.: The data acquisition for the first half, 1-6 hours (blue), and the second half, 7-12 hours (orange), is plotted in the graph. The y-axis indicates power (gain) in arbitrary units. The minimum difference in the noise floor in two readings is approx 0.03% while the maximum difference is approx. 0.3%[26].

The detection pipeline should be robust against noise fluctuations, and the impact of these fluctuations on the detection efficiency should be as small as possible.

To mimic the noise fluctuations, we added and subtracted some offset to the noise data in the frequency domain and examined its effects on the False positive rate of the AE and the baseline. Matched filtering is not considered for this test because the applied noise offset changes only the mean noise level, while the matched-filter response is determined by the correlation with the known signal template.

4.3.1. Offset Added

In this section, we will examine the effects of an added offset to the noise in the frequency domain. For simplicity, only a constant noise offset is added.

The offset fraction was varied from 0.03% to 10%. The noise offset is calculated as

$$\text{Constant offset} = k \times < Noise >, \tag{4.2}$$

where k is the offset fraction (i.e. 0.03% to 10%) and $< Noise >$ is the spectral average of noise in the frequency domain for each noise file.

This constant offset is added to the entire noise file in the frequency domain. The resulting FPR is then recalculated while keeping the thresholds determined in Section 3.3.3 unchanged. Throughout this process, AE's training and the average spectral noise estimate used for baseline detections remains unchanged.

To re-calculate the FPR, first reconstruction loss for offset added noise data is calculated. Then the probability density distribution is extrapolated and FPR is determined corresponding to the thresholds given in Table 3.1.

The resulting change in the FPR from the initially chosen value of 1/year is shown in the table 4.1.

Offset (%)	**Baseline FPR(per year)**	**AE FPR (per year)**
0.03	1.003 ± 6.662	0.842 ± 2.262
0.06	1.013 ± 6.726	0.847 ± 2.273
0.10	1.009 ± 6.705	0.852 ± 2.286
1.00	1.224 ± 8.047	0.972 ± 2.61
10.00	24.579 ± 129.705	3.967 ± 10.625

Table 4.1.: False positive rate per year for different positive offset values with their respective uncertainties.

The uncertainty in the FPR is calculated by using error propagation of the uncertainty of thresholds. From table 4.1, it can be concluded that AE shows a lesser deviation in FPR over the full range of offset values in comparison to the baseline method. When a small offset, up to 0.1% is added, both methods are comparable in performance, but the baseline shows higher uncertainty than AE, making AE the more robust of the two. When an offset of 1% or higher is added to the noise, the baseline method shows a higher deviation in FPR with a significantly higher uncertainty. On the other hand, at 1% added offset, AE shows minimal FPR deviation with a smaller uncertainty than the excess power method. Even at 10% offset added, the AE has an FPR of 3.87 ± 12.37 per year. This shows that the AE remains relatively robust under a large offset. In comparison, the baseline method yields a much higher FPR of 24.579 /yr, with a very large uncertainty of 129.705/yr.

These results clearly show that the AE performs better in the presence of a constant positive noise offset. In a practical scenario, AE's robustness and stability against the noise fluctuations over time will be highly valuable for the experiment.

4.3.2. Offset Subtracted

In this section, we will subtract a constant offset from the noise data in the frequency domain and examine how detection efficiency is affected.

The offset is calculated with equation 4.2, and the offset fraction is varied from 0.03% to 10%.

It should be noted that during this analysis, the threshold is kept fixed, though the corresponding FPR may change due to the presence of a negative offset in the data. This is done to mimic the practical scenario where the detector is not calibrated over a short duration.

In Figure 4.4, Tables 4.3 and 4.2, the relative difference in SNR to achieve 95% detection efficiency can be observed for AE and baseline, respectively.

The relative difference in efficiency is calculated as

$$\Delta R = \frac{2\ |A - B|}{A + B} \times 100, \tag{4.3}$$

where A denotes the SNR required to achieve 95% detection efficiency in the absence of any noise offset in the data, and B denotes the corresponding SNR in the presence of a negative noise offset.

The uncertainty in the relative difference is calculated by

$$\begin{aligned} R_{\sigma_+} &= \frac{4}{(A+B)^2}\sqrt{B^2\,(\sigma_A^+)^2 + A^2\,(\sigma_B^-)^2} \times 100, \\ R_{\sigma_-} &= \frac{4}{(A+B)^2}\sqrt{(B^2\,(\sigma_A^-)^2 + A^2\,(\sigma_B^+)^2} \times 100. \end{aligned} \tag{4.4}$$

In Equation 4.4, σ_A^+ and σ_A^- denote the upper and lower uncertainties in A, respectively, while σ_B^+ and σ_B^- denote the corresponding upper and lower uncertainties in B.

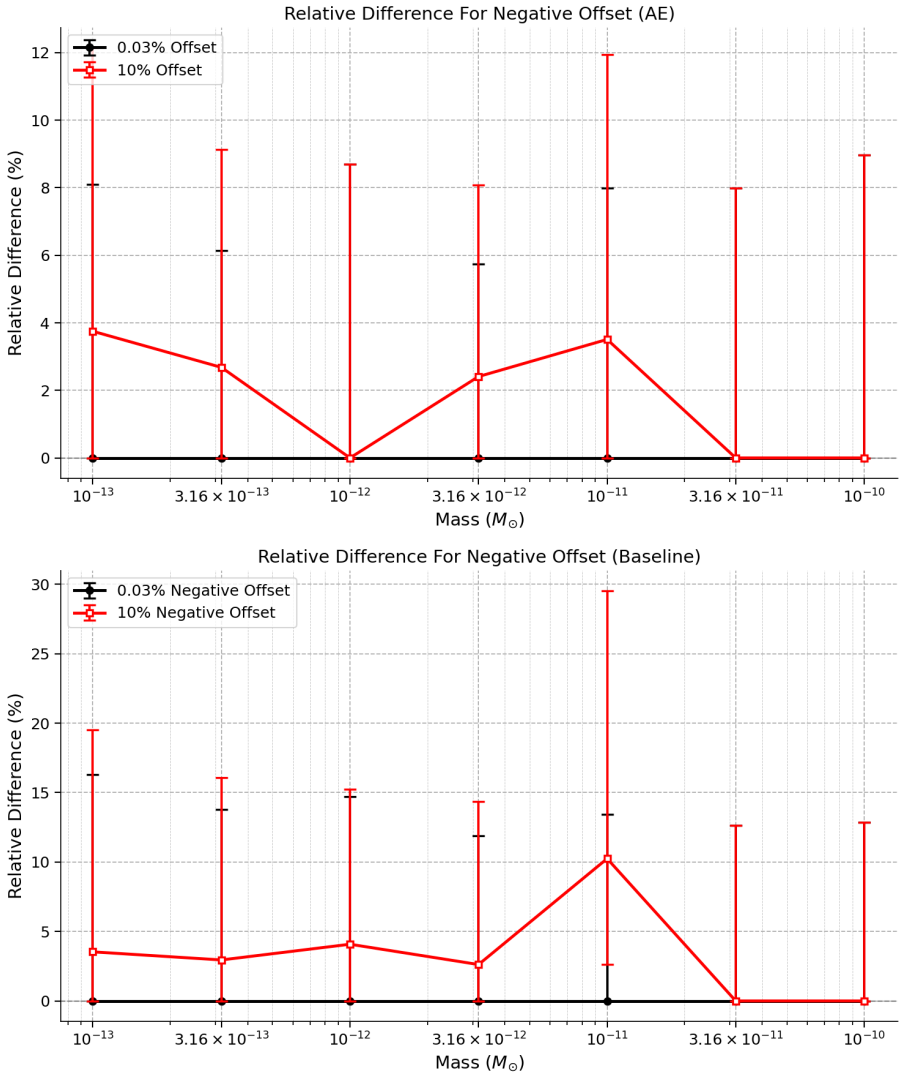


Figure 4.4.: Relative difference between the SNR at 95% detection efficiency with a negative offset and without a negative offset. The image on top shows the relative difference in percentage for AE, and the image at the bottom shows the percentage relative difference for baseline.

Both AE and baseline are stable when negative offsets of 0.03% to 1% are present in the data, and the SNR required for 95% detection efficiency stays unchanged. When a constant offset with a 10% fraction is subtracted from the noise, the SNR required for 95% detection efficiency increases for both of the detection pipelines.

For AE, the highest relative difference is observed for the signal corresponding to the PBH mass of $10^{-11}M_\odot$ with a relative difference of 3.57%. For signals corresponding to the PBH masses of $10^{-10}M_\odot$ and $3.162 \times 10^{-10}M_\odot$ there is no change in SNR for 95 %

detection even at 10% of negative offset. The notable relative difference is only seen in lighter mass PBH merger signals and with an offset fraction of 10% . Overall, AE is stable over constant negative offsets.

Baseline shows a higher relative difference in SNR for 95% signal detection when 10% constant offset is present. The highest relative difference is for the signal corresponding to the PBH mass of $10^{-11} M_{\odot}$ with 10.80%. Similar to the AE, the change in SNR for 95% detection efficiency is only observed for the offset of 10%. Overall, the baseline method shows a higher relative difference percentage compared to AE. Thus, AE is more stable when a negative offset is present in the data.

Table 4.2.: Relative difference in the baseline method's SNR required for 95% detection efficiency, comparing data with and without noise-offset subtraction. For each offset fraction, the relative difference is reported with its corresponding uncertainty.

Mass ($M_{\odot}$)	**Offset fraction**				
	0.03%	**0.06%**	**0.1%**	**1%**	**10%**
1×10^{-10}	$0.00^{+12.86}_{-0.00}$	$0.00^{+12.86}_{-0.00}$	$0.00^{+12.86}_{-0.00}$	$0.00^{+12.86}_{-0.00}$	$0.00^{+12.86}_{-0.00}$
3.162×10^{-11}	$0.00^{+12.65}_{-0.00}$	$0.00^{+12.65}_{-0.00}$	$0.00^{+12.65}_{-0.00}$	$0.00^{+12.65}_{-0.00}$	$0.00^{+12.65}_{-0.00}$
1×10^{-11}	$0.00^{+13.42}_{-0.00}$	$0.00^{+13.42}_{-0.00}$	$0.00^{+13.42}_{-0.00}$	$0.00^{+13.42}_{-0.00}$	$10.25^{+19.29}_{-7.60}$
3.162×10^{-12}	$0.00^{+11.91}_{-0.00}$	$0.00^{+11.91}_{-0.00}$	$0.00^{+11.91}_{-0.00}$	$0.00^{+11.91}_{-0.00}$	$2.62^{+11.73}_{-2.62}$
1×10^{-12}	$0.00^{+14.73}_{-0.00}$	$0.00^{+14.73}_{-0.00}$	$0.00^{+14.73}_{-0.00}$	$0.00^{+14.73}_{-0.00}$	$4.08^{+11.15}_{-4.08}$
3.162×10^{-13}	$0.00^{+13.78}_{-0.00}$	$0.00^{+13.78}_{-0.00}$	$0.00^{+13.78}_{-0.00}$	$0.00^{+13.78}_{-0.00}$	$2.95^{+13.14}_{-2.95}$
1×10^{-13}	$0.00^{+16.31}_{-0.00}$	$0.00^{+16.31}_{-0.00}$	$0.00^{+16.31}_{-0.00}$	$0.00^{+16.31}_{-0.00}$	$3.53^{+15.99}_{-3.53}$

Table 4.3.: Relative difference in the AE's SNR required for 95% detection efficiency, comparing data with and without noise-offset subtraction. For each offset fraction, the relative difference is reported with its corresponding uncertainty.

Mass ($M_\odot$)	**Offset fraction**				
	0.03%	**0.06%**	**0.1%**	**1%**	**10%**
1×10^{-10}	$0.00^{+8.98}_{-0.00}$	$0.00^{+8.98}_{-0.00}$	$0.00^{+8.98}_{-0.00}$	$0.00^{+8.98}_{-0.00}$	$0.00^{+8.98}_{-0.00}$
3.162×10^{-11}	$0.00^{+7.99}_{-0.00}$	$0.00^{+7.99}_{-0.00}$	$0.00^{+7.99}_{-0.00}$	$0.00^{+7.99}_{-0.00}$	$0.00^{+7.99}_{-0.00}$
1×10^{-11}	$0.00^{+7.99}_{-0.00}$	$0.00^{+7.99}_{-0.00}$	$0.00^{+7.99}_{-0.00}$	$0.00^{+7.99}_{-0.00}$	$3.51^{+8.43}_{-3.51}$
3.162×10^{-12}	$0.00^{+5.75}_{-0.00}$	$0.00^{+5.75}_{-0.00}$	$0.00^{+5.75}_{-0.00}$	$0.00^{+5.75}_{-0.00}$	$2.41^{+5.68}_{-2.41}$
1×10^{-12}	$0.00^{+8.70}_{-0.00}$	$0.00^{+8.70}_{-0.00}$	$0.00^{+8.70}_{-0.00}$	$0.00^{+8.70}_{-0.00}$	$0.00^{+8.70}_{-0.00}$
3.162×10^{-13}	$0.00^{+6.15}_{-0.00}$	$0.00^{+6.15}_{-0.00}$	$0.00^{+6.15}_{-0.00}$	$0.00^{+6.15}_{-0.00}$	$2.68^{+6.45}_{-2.68}$
1×10^{-13}	$0.00^{+8.11}_{-0.00}$	$0.00^{+8.11}_{-0.00}$	$0.00^{+8.11}_{-0.00}$	$0.00^{+8.11}_{-0.00}$	$3.75^{+8.39}_{-3.75}$

5. Discussion

In this thesis, we carefully built and evaluated 2 ML-based detection pipelines. Both of the pipelines were compared with 3 different methods of signal detection on various aspects.

Each method tested in this thesis has a unique advantage over all others. Starting with AE, it is the most robust method of all for handling offsets in noise data. AE maintains a stable FPR even when a constant noise offset is added to the data. While the baseline method reaches the 95% detection efficiency at the lowest SNRs in the component mass range of $10^{-13}M_{\odot}$ to $3.162 \times 10^{-11}M_{\odot}$. However, the AE results for detector efficiency are within the uncertainty range. The time domain CNN LSTM classifier shows a lower SNR for 95% detection efficiency corresponding to the PBH mass of $10^{-10}M_{\odot}$. The AE's performance at this specific data point might be a result of an artifact introduced during signal generation. As the CNN LSTM classifier uses a different signal generation pipeline, it is unaffected by this effect.

As said earlier, the baseline method also detects unknown signal shapes from noisy data without a known signal waveform, but it shows lower tolerance to noise offsets. This result is particularly important as in practical scenarios, the baseline method would require more frequent calibrations, and detection sensitivity could be significantly lowered in the case of sudden offset addition. The baseline method is also less stable than AE when a negative offset is present in the data.

Excess power method consumes the least amount of resources in pre-processing. Compared to AE's training pipeline, the baseline needs only noise average calculation and threshold calculation based on the residuals. This is a clear advantage for the baseline method as it eliminates any requirement for lengthy training, but it comes at the cost of lower stability against noise offsets.

Signals corresponding to the component mass range of $3.162 \times 10^{-11}M_{\odot}$ to $10^{-10}M_{\odot}$ were tested with matched filtering technique. The signal template corresponding to the PBH mass of $10^{-11}M_{\odot}$ was used for the same. The SNR required to achieve 95% detection efficiency is approximately 2-3 for all tested signals. This performance is better than AE and baseline for signals corresponding to masses of $3.162 \times 10^{-11}M_{\odot}$ and $10^{-10}M_{\odot}$ but significantly poor for the signal corresponding to the PBH mass of $3.162 \times 10^{-12}M_{\odot}$. For the signal perfectly matching the template at the PBH mass of $10^{-11}M_{\odot}$ the performance of matched filtering technique is very similar to AE and baseline. As matched filtering uses the optimal filter for this waveform, this mass point is expected to represent its best performance among all tested signals. Furthermore, matched filtering outperforms the CNN-LSTM classifier, the other time-domain method evaluated in this thesis, across all tested data points. The matched filtering method was not tested against the noise offsets

as only constant offset is added to the noise data and theoretically it won't change the matched filter response determined by the correlation with the known signal template.

It should be noted that the Matched filtering method requires a signal template; thus, any signal of unknown waveform would be missed. In this thesis, only the inspiral phase of the merger is taken into account while simulating the signals, and this is one of the reasons we could use only one template for signals with different sweep rates. Another reason contributing to it is the multiplication of noisy data with the complex conjugate of the signal template in the frequency domain. Due to this step, the phase information is lost, and one template is able to extract signals corresponding to different PBH masses. is In practical cases, a signal of a different shape or in different stages of merger for the same component mass could be easily missed. Moreover, to execute a matched filtering based detection pipeline, a huge template bank would be required. This makes the matched filtering method most complex one in terms of pre-processing. Despite being robust against many offsets, this method is most difficult to apply to GravNet's detection pipeline.

The classifier detection pipeline has a few constraints due to the limited availability of the data, so we compared the results with other methods at a higher FPR of 8.869×10^{10}/ yr. This very obstacle could also be there during observation. Due to the limited extrapolation possibility of the classifier's output, it will always have a higher FPR than other methods. Though we can proceed with a higher FPR as long as we have the capacity to further analyze the flagged input vector.

Compared to AE classifier requires more amount of data and training time. Despite this the performance remains very comparable to AE.

Due to the limitation of resources, such as the limited availability of the noise data, no concrete conclusion can be made for the classifier's performance.

Lastly, we will discuss the time domain based ML detection pipeline, the CNN LSTM classifier[24]. The pipeline requires the highest SNR to achieve 95% detection efficiency for the signals corresponding to the PBH mass range of $10^{-13}M_{\odot}$ to $10^{-11}M_{\odot}$. The reason behind it could be the advanced pattern recognition possibility in the frequency domain, which allows the tested AE to detect signals with lower SNR at the same detection efficiency. For the signals corresponding to the masses $3.162 \times 10^{-11}M_{\odot}$ and $10^{-10}M_{\odot}$ CNN LSTM classifier requires significantly lower SNR than the frequency domain detection pipelines like AE and baseline technique. As discussed earlier this might be due to an artifact present in simulated signals used to test the AE ans baseline while the CNN LSTM classifier uses different simulated signals which are not affected by the similar artifact.

Overall, the AE performs well across all evaluation criteria considered in this thesis. It is robust against noise offsets, and the SNR required to achieve 95% detection efficiency is one of the lowest for all tested methods. These results indicate that AE-based and, more broadly, machine-learning-based detection pipelines are promising candidates for gravitational-wave detection, and that further research in this direction would be beneficial.

6. Conclusion

6.1. Summary

In this thesis, machine learning based detection pipelines for high-frequency gravitational waves detection were developed and evaluated on various criteria.

To test these detection pipelines, high-frequency gravitational wave signals were simulated. The simulated signals correspond to the inspiral phase of an equal mass binary PBH merger. For simplicity, only linear chirps with constant amplitude were considered for the simulation. Once the signals were generated they were carefully injected in the noise data obtained from the SUPAX cavity at Mainz.

First, a CNN Autoencoder was designed for anomaly detection. The AE was trained on the frequency domain pure noise data from the SUPAX cavity at Mainz. It is designed to reconstruct the input as closely as possible at the output. The difference between the input and the output is called the 'reconstruction loss'. The higher the reconstruction loss, the more the probability of having an anomaly (in this case, a signal) in the given input vector.

When a fully trained AE is given an input vector with a signal embedded in noise, it fails to reconstruct the signal waveform and only reconstructs the noise shape. As a result, a high reconstruction loss is generated, separating the input vector from noise only data.

To regulate the anomaly detection with the desired False Positive Rate (FPR), a suitable threshold on reconstruction loss was chosen.

To further enhance the detection sensitivity, we designed a multi-view classifier. The classifier was trained on 3 views of the input data, namely 1) signal + noise / pure noise data (50/50 distribution), 2) reconstruction from AE, and 3) reconstruction loss from AE. The classifier inputs were labeled as signal containing (y=1) and pure noise (y=0). The model is trained in a supervised manner and outputs the positive class probability, which is compared with the class label by using the binary cross-entropy loss.

To obtain an output class label, a threshold is chosen on the output of the classifier. The threshold is determined from the receiver operating characteristic curve and a desired FPR.

Lastly, the excess power method was used as a baseline to compare the detection efficiency of both ML based pipelines. This method compares each test sample with the average spectral noise shape and uses the summed residual power as a detection statistic.

All three techniques were compared against parameters such as detection efficiency, constant offset addition, and subtraction.

Additionally, the results from the time domain CNN LSTM classifier and matched filtering technique were also compared with the ML-based pipelines.

The comparison shows that each method has distinct advantages and limitations. The autoencoder provides the most robust performance against noise-floor variations. Since it is trained only on noise data, it does not require prior knowledge of the signal waveform and is therefore well suited for weakly modeled or unknown transient signals. Its detection efficiency is slightly lower than that of the baseline method, but the results remain comparable within uncertainties.

The excess-power baseline method reaches 95% detection efficiency at the lowest SNR for PBH mass of $10^{-13}M_{\odot}$ to $10^{-11}M_{\odot}$ and AE is a close second in this range. But baseline is computationally much cheaper. However, the method is more sensitive to offsets in the noise data and might need constant re-callibrations.

Matched Filtering was tested for signals corresponding to the PBH mass range of $3.162 \times 10^{-12}M_{\odot}$ to $10^{-10}M_{\odot}$ and only one signal template was used, corresponding to the PBH mass of $10^{-11}M_{\odot}$. The perfomance of matched filtering is comparable to baseline and AE at data point corresponding to $10^{-11}M_{\odot}$ where the same signal template is 100% match. But for heavier PBH masses the matched filtering shows better results than the frequency domain detection pipelines. This might be due to the artifact present in the signal embedded data. For signals corresponding to $3.162 \times 10^{-12}M_{\odot}$, AE and baseline both achieve 95% detection efficiency at much lower SNR than matched filtering. Overall matched filtering gives mixed results and only works with a prior know signal waveform.

The CNN LSTM classifier needs much higher SNR to attain 95% detection efficiency than frequency domain methods at data points corresponding to $10^{-13}M_{\odot}$ to $10^{-11}M_{\odot}$. but performs better than frequency domain methods for signals corresponding to PBH masses of $3.162 \times 10^{-11}M_{\odot}$ to $10^{-10}M_{\odot}$ where an artifact is present in simulated signals used to test the frequency domain pipelines, but is absent from the signal simulatons used to test the CNN LSTM classifier. Despite being a time domain based supervised neural network, requires higher SNR for 95% detection efficiency than matched filtering.

Classifier was tested on much higher FPR of 8.869×10^{10}/ yr due to the data limitations. When AE ans baseline are compared with classifier with the same FPR , it can be seen that classifier performance is very similar to to both AE and baseline methods. The SNR required to achieve 95 % detection efficiency is similar to these two methods despite a much more training needs of the classifier.

Overall, the results indicate that frequency-domain machine-learning methods, especially the autoencoder-based pipeline, provide a promising complementary approach for the detection of weakly modeled transient HFGW signals.

A. Code Listings

The code used in this thesis can be found on this link

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Note:AI-based tools such as Grammerly, ChatGpt and Claude were used for spelling and grammar check in this thesis.